

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

Submitted solver team

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Abstract

This report is generated from completed solver outputs only. It covers all eight required cases, their submitted convergence status, force and residual evidence, field figures, MPI partition evidence, rank-count comparisons, and the cylinder Reynolds-200 shedding analysis. The automation rejects failed, incomplete, or internally inconsistent report inputs; it does not independently establish physical correctness or solver provenance.

1 Introduction

The benchmark solves the two-dimensional NACA0012 and circular-cylinder cases with a cell-centred, unstructured finite-volume compressible-flow solver. Required results are accepted here only when their submitted status is `converged` or `statistically_periodic`. Values in tables and figures are read from the submitted CSV, VTK, JSON, and partition-diagnostic files; no histories, fields, or convergence labels are synthesized by this script. These checks establish output consistency, not independent physical validation or proof of solver provenance.

2 Governing equations and nondimensionalization

The conservative state is $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ and the reported equation set is the two-dimensional compressible Navier–Stokes system. The discretized equation is

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \quad \mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

For the calorically perfect gas, $p = (\gamma - 1)\rho e$, $E = e + \frac{1}{2}(u^2 + v^2)$, and $a = \sqrt{\gamma p / \rho}$. The supplied inputs use $\gamma = 1.4$, $R = 1$, and $Pr = 0.72$; laminar cases use constant viscosity $\mu = \rho_\infty U_\infty L_{ref} / Re$. With $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$, the reported coefficients are $C_D = D/(q_\infty A_{ref})$, $C_L = L/(q_\infty A_{ref})$, and $C_m = M_z/(q_\infty A_{ref} L_{ref})$.

For viscous fluxes the Newtonian/Fourier model is $\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y)$, $\tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y)$, $\tau_{xy} = \mu(u_y + v_x)$, and $\mathbf{q} = -\mu c_p \nabla T / Pr$. The force split reports pressure traction separately from the tangential component of viscous traction; C_f is based on this tangential traction rather than normal viscous traction.

3 Meshes, boundaries, and spatial discretization

Each residual is a sum of oriented face fluxes over an unstructured control volume:

$$R_i = \sum_{f \in \partial\Omega_i} (\hat{F}_f^i - \hat{F}_f^v) |S_f|.$$

Each face normal is directed from the owner cell to its neighbour or exterior. The submitted metadata records the case-selectable inviscid flux: inviscid NACA packages use Rusanov variants, with recorded total-enthalpy consistency and rarefaction protection where active, while the remaining cases use HLLC with an admissibility-checked Rusanov fallback. The case inputs map CGNS boundary families to farfield plus either inviscid slip wall or viscous no-slip adiabatic wall. Farfield faces use a characteristic exterior state. Every stationary impermeable body face uses the exact inviscid flux $[0, pn_x, pn_y, 0]^T$ rather than solving a reflected reconstructed wall Riemann problem; no-slip walls additionally impose $u = v = 0$ and zero normal temperature gradient in the viscous flux. Submitted surface data are declared as `boundary_value` output, so wall velocities in the report are not conflated with adjacent cell-centre velocities.

Gradients are weighted least-squares reconstructions and face states are piecewise linear. The production limiter is the submitted Barth–Jespersen method, followed by a pressure-jump sensor that smoothly flattens all primitive gradients only across captured shocks while remaining exactly inactive below its smooth-region onset. Exact linear reproduction is unit-tested on an irregular stencil, and an asymptotic scaled irregular-stencil manufactured-solution regression requires observed smooth face-value order greater than 1.95 across successive refinements with the production sensor inactive. Face-state density/pressure positivity uses the submitted scaling/backtracking method. The report does not infer any other first-order fallback: its occurrence and trigger must be recorded in run notes or metadata before it can be claimed.

Table 1: Submitted mesh/case evidence. Boundary tags are the tags present in submitted wall surface rows.

Case	cells	faces	Mach	mode	surface tags
naca0012_m015_inviscid	20816	36498	0.15	inviscid	bc-4
naca0012_m080_inviscid	20816	36498	0.8	inviscid	bc-4
naca0012_m200_inviscid	20816	36498	2	inviscid	bc-4
naca0012_m015_laminar_re5000	20816	36498	0.15	laminar	bc-4
naca0012_m080_laminar_re5000	20816	36498	0.8	laminar	bc-4
naca0012_m200_laminar_re5000	20816	36498	2	laminar	bc-4
cylinder_m010_laminar_re20	10185	20180	0.1	laminar	WALL
cylinder_m010_laminar_re200	10185	20180	0.1	laminar	WALL

The production metadata identifies the reconstruction, limiter, and positivity method for every result: piecewise-linear reconstruction is applied to face states, the stated limiter controls oscillations, and the stated positivity control protects density and pressure. The report does not infer an unused first-order fallback; any fallback must be described by the submitted metadata/notes.

4 Implicit steady march and true transient BDF2

Steady cases use the submitted local pseudo-time integrator and inner solver. They retain the supplied startup CFL and ramp horizon while applying a documented conservative terminal cap; Anderson history is reset whenever the CFL changes. Accelerated iterates must strictly reduce both global L_2 and L_∞ residuals, and target convergence requires both norms plus a stable force tail.

Table 2: Production numerical methods. Concise labels are used for readability; exact method identifiers are retained in each `metadata.json`.

Case	inviscid flux	viscous flux	time integrator	implicit solver
NACA0012 M=0.15 inviscid	Rusanov	none	local pseudo-time	4×4 block Jacobi
NACA0012 M=0.8 inviscid	Rusanov (H_0 consistent)	none	local pseudo-time	4×4 block Jacobi
NACA0012 M=2 inviscid	Rusanov (H_0 consistent, rarefaction guard)	none	local pseudo-time	4×4 block Jacobi
NACA0012 M=0.15 Re=5000	HLLC + Rusanov fallback	Newtonian/Fourier	local pseudo-time	4×4 block Jacobi
NACA0012 M=0.8 Re=5000	HLLC + Rusanov fallback	Newtonian/Fourier	local pseudo-time	4×4 block Jacobi
NACA0012 M=2 Re=5000	HLLC + Rusanov fallback	Newtonian/Fourier	local pseudo-time	4×4 block Jacobi
Cylinder M=0.1 Re=20	HLLC + Rusanov fallback	Newtonian/Fourier	local pseudo-time	4×4 block Jacobi
Cylinder M=0.1 Re=200	HLLC + Rusanov fallback	Newtonian/Fourier	dual-time BDF2	4×4 block Jacobi

The stationary-wall block uses the analytic pressure-flux Jacobian while its spectral radius remains in the pseudo-time mass. Requested controls and detectable departures are listed in table 3. For the Re 200 cylinder, metadata identifies `true_dual_time_BDF2`, with true BDF2 inner loop set to `True`. The required outer physical-time loop holds U^n and U^{n-1} fixed while inner iterations solve for U^{n+1} , then advances the histories after accepted inner convergence. Its submitted observed inner-iteration range is 133–283, mean 210.66, target-miss count 0, converged-step fraction 1, and last ratio 0.00098623.

5 MPI decomposition, halo exchange, and no replication

The metadata identifies the partitioner and halo exchange for every submitted run. It explicitly declares both full-mesh and full-conservative-state replication during solver iterations false. Partition diagnostics below are the actual rank-local owned/ghost counts, neighbour ranges, and edge-cut metadata, rather than a preprocessing-only estimate. Halo communication is neighbour-scoped as named in metadata; residuals and forces are the submitted global histories.

6 Output reproducibility and sanity checks

The generated `run_manifest.csv` records both production and rank-validation commands, source directories, ranks, wall times, final steps, and statuses. `figure_manifest.csv` maps every figure to the actual CSV or field file and named variable. `sanity_checks.json` records contract-derived positivity, wall-condition, force, pressure-coefficient, and Re 200 wake checks. Regeneration is documented with this repository’s report tool; the TeX source is ready for `latexmk`.

Table 3: Supplied and observed controls. Steady ramp details are explicitly marked unavailable when metadata does not store them.

Case	supplied controls	actual stored evidence	detected deviation
naca0012_m015_inviscid	N=20000; R=4; CFL=1–100; ramp=2000; I=3–50; r=0.01	CFL=1–10; N=4782; I=3–50; r=0.01; ramp=observed directly in residual CFL rows	safeguarded terminal CFL 10 is below the supplied cap 100
naca0012_m080_inviscid	N=30000; R=4; CFL=1–100; ramp=3000; I=3–50; r=0.01	CFL=1–2; N=3942; I=3–3; r=0.01; ramp=observed directly in residual CFL rows	safeguarded terminal CFL 2 is below the supplied cap 100
naca0012_m200_inviscid	N=40000; R=3; CFL=0.2–50; ramp=5000; I=3–80; r=0.01	CFL=0.2–0.40899; N=2223; I=3–3; r=0.01; ramp=observed directly in residual CFL rows	safeguarded terminal CFL 0.408992 is below the supplied cap 50
naca0012_m015_laminar_re5000	N=40000; R=4; CFL=1–100; ramp=5000; I=3–80; r=0.01	CFL=1–5; N=5145; I=4–23; r=0.01; ramp=observed directly in residual CFL rows	safeguarded terminal CFL 5 is below the supplied cap 100
naca0012_m080_laminar_re5000	N=40000; R=4; CFL=0.5–80; ramp=5000; I=3–80; r=0.01	CFL=0.5–1; N=5000; I=5–8; r=0.01; ramp=observed directly in residual CFL rows	safeguarded terminal CFL 1 is below the supplied cap 80
naca0012_m200_laminar_re5000	N=50000; R=3; CFL=0.2–50; ramp=7000; I=3–100; r=0.01	CFL=0.2–0.5723; N=4573; I=3–5; r=0.01; ramp=observed directly in residual CFL rows	safeguarded terminal CFL 0.572298 is below the supplied cap 50
cylinder_m010_laminar_re20	N=30000; R=5; CFL=1–100; ramp=3000; I=3–80; r=0.01	CFL=1–5; N=3716; I=5–26; r=0.01; ramp=observed directly in residual CFL rows	safeguarded terminal CFL 5 is below the supplied cap 100
cylinder_m010_laminar_re200	dt=0.01; tf=300; I=5–1000; r=0.001	CFL=1–1; N=30000; I=133–283; r=0.001; dt observed 0.01–0.01; tf 300.0	none detected from stored evidence

Table 4: Final-run partition evidence. Owned, ghost, and neighbour entries are rankwise min–max; balance is maximum owned cells divided by the mean.

Case	ranks	owned	ghost	neighbours	balance	cut
naca0012_m015_inviscid	8	2590–2619	60–177	2–6	1.007	447
naca0012_m080_inviscid	8	2590–2619	60–177	2–6	1.007	447
naca0012_m200_inviscid	8	2590–2619	60–177	2–6	1.007	447
naca0012_m015_laminar_re5000	8	2590–2619	60–177	2–6	1.007	447
naca0012_m080_laminar_re5000	8	2590–2619	60–177	2–6	1.007	447
naca0012_m200_laminar_re5000	8	2590–2619	60–177	2–6	1.007	447
cylinder_m010_laminar_re20	8	1267–1279	96–133	3–7	1.005	483
cylinder_m010_laminar_re200	8	1267–1279	96–133	3–7	1.005	483

Table 5: Submitted run status. Both residual reductions are recomputed from the first and final rows.

Case	ranks	steps	time	L_2 orders	L_∞ orders	wall s	status
NACA0012 M=0.15 inviscid	8	4782	0	5.33	5.13	1164	converged
NACA0012 M=0.8 inviscid	8	3942	0	0.569	0.856	531.4	converged
NACA0012 M=2 inviscid	8	2223	0	3	3.26	324.9	converged
NACA0012 M=0.15 Re=5000	8	5145	0	4.34	4	2670	converged
NACA0012 M=0.8 Re=5000	8	5000	0	0.371	0.35	1187	converged
NACA0012 M=2 Re=5000	8	4573	0	3.65	3.56	1118	converged
Cylinder M=0.1 Re=20	8	3716	0	5.01	5.01	1073	converged
Cylinder M=0.1 Re=200	8	30000	300	1.29	1.19	1.154e+04	stat. periodic

Table 6: Final force coefficients from each submitted force history.

Case	C_D	C_L	C_m	pressure drag	viscous drag
naca0012_m015_inviscid	0.00056425	-1.5125e-05	4.5244e-05	0.00056425	0
naca0012_m080_inviscid	0.020114	-0.0015247	0.00028173	0.020114	0
naca0012_m200_inviscid	0.091753	-0.0014795	-0.0003376	0.091753	0
naca0012_m015_laminar_re5000	0.056405	-0.00041188	8.4649e-06	0.018391	0.038013
naca0012_m080_laminar_re5000	0.0889	-0.00042574	8.1395e-05	0.048752	0.040148
naca0012_m200_laminar_re5000	0.11573	0.00013323	-4.9751e-05	0.010916	0.10482
cylinder_m010_laminar_re20	2.049	0.00063295	2.8194e-06	1.2351	0.8139
cylinder_m010_laminar_re200	1.2747	-0.52747	0.0044818	1.0319	0.24275

7 Results

7.1 Run status and final forces

7.2 naca0012_m015_inviscid

The submitted status is **converged**; the computed first-to-last L_2/L_∞ residual reductions are 5.33/5.13 orders. The final force row gives $C_D = 0.00056425$, $C_L = -1.5125e - 05$, and $C_m = 4.5244e - 05$. Figure 1, Figure 2, Figure 3, Figure 4, Figure 5 provide the residual, force, wall C_p/C_f , Mach, and pressure evidence.

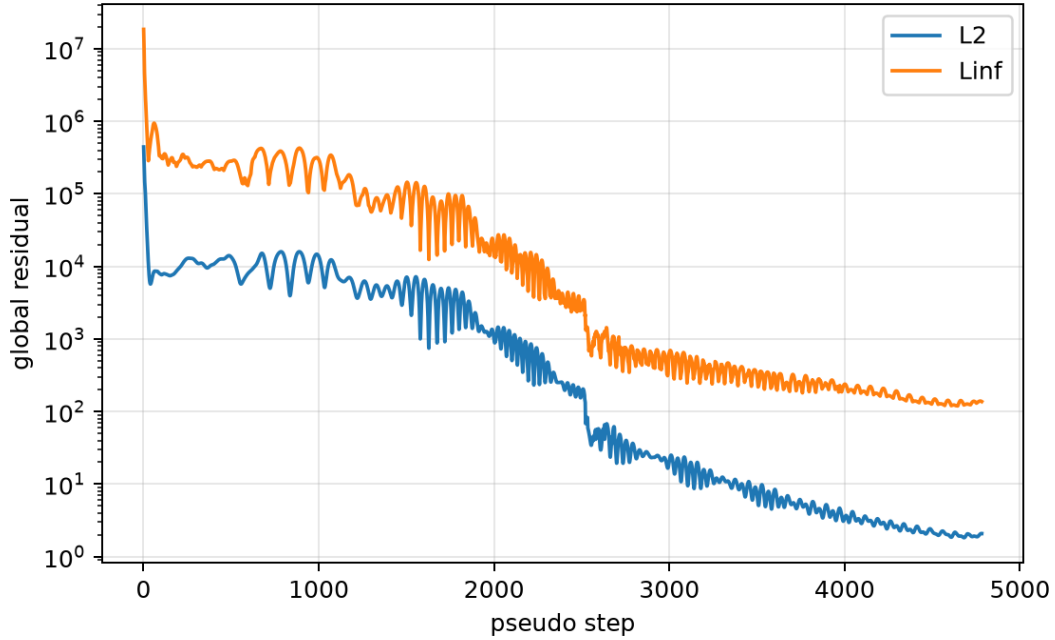


Figure 1: Global residual history.

7.3 naca0012_m080_inviscid

The submitted status is **converged**; the computed first-to-last L_2/L_∞ residual reductions are 0.569/0.856 orders. The final force row gives $C_D = 0.020114$, $C_L = -0.0015247$, and $C_m =$

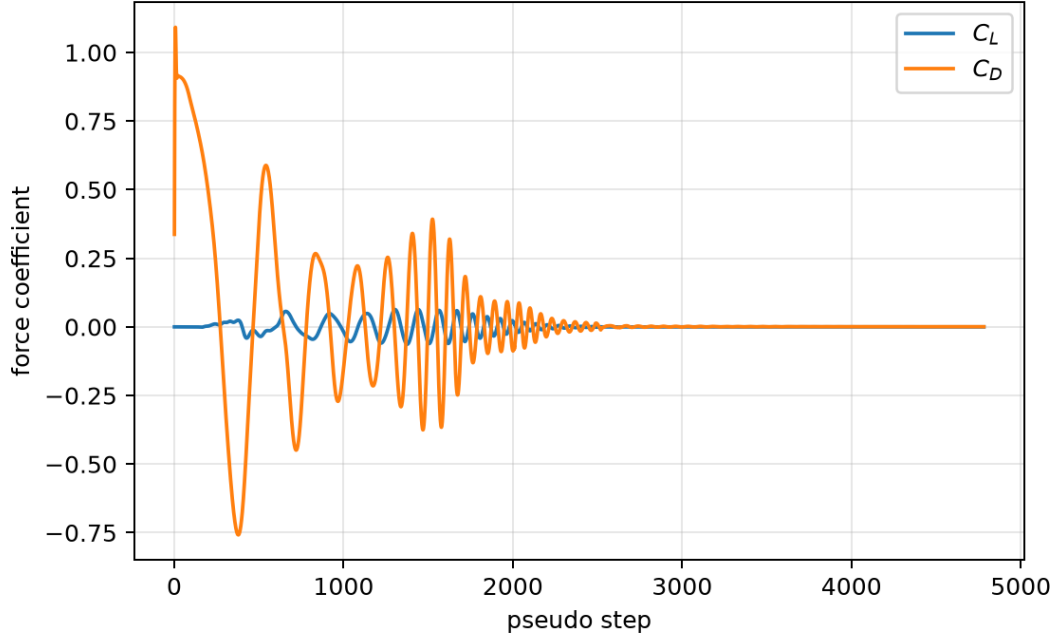


Figure 2: Force-coefficient history.

0.00028173. Figure 6, Figure 7, Figure 8, Figure 9, Figure 10 provide the residual, force, wall C_p/C_f , Mach, and pressure evidence.

7.4 naca0012_m200_inviscid

The submitted status is **converged**; the computed first-to-last L_2/L_∞ residual reductions are 3/3.26 orders. The final force row gives $C_D = 0.091753$, $C_L = -0.0014795$, and $C_m = -0.0003376$. Figure 11, Figure 12, Figure 13, Figure 14, Figure 15 provide the residual, force, wall C_p/C_f , Mach, and pressure evidence.

7.5 naca0012_m015_laminar_re5000

The submitted status is **converged**; the computed first-to-last L_2/L_∞ residual reductions are 4.34/4 orders. The final force row gives $C_D = 0.056405$, $C_L = -0.00041188$, and $C_m = 8.4649e - 06$. Figure 16, Figure 17, Figure 18, Figure 19, Figure 20 provide the residual, force, wall C_p/C_f , Mach, and pressure evidence.

7.6 naca0012_m080_laminar_re5000

The submitted status is **converged**; the computed first-to-last L_2/L_∞ residual reductions are 0.371/0.35 orders. The final force row gives $C_D = 0.0889$, $C_L = -0.00042574$, and $C_m = 8.1395e - 05$. Figure 21, Figure 22, Figure 23, Figure 24, Figure 25 provide the residual, force, wall C_p/C_f , Mach, and pressure evidence.

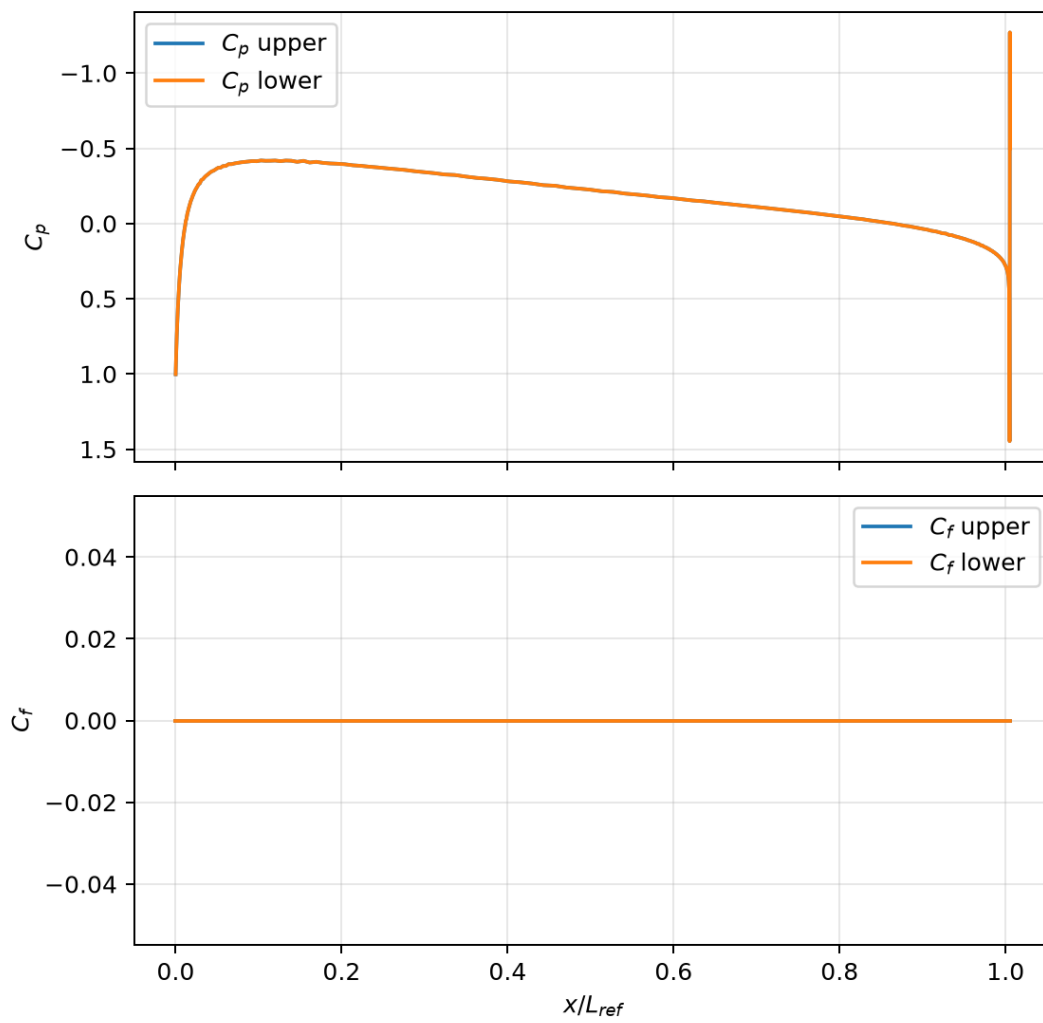


Figure 3: Submitted surface pressure and skin-friction coefficients.

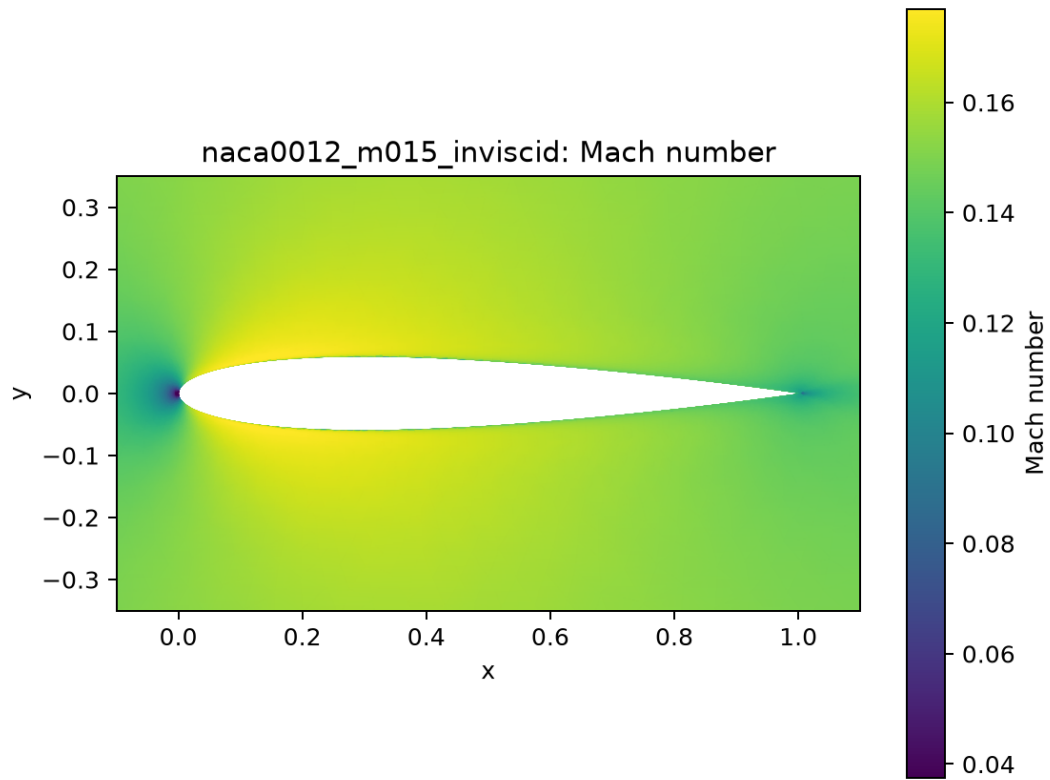


Figure 4: Mach number from the final submitted field; the displayed percentile-clipped range is $[0.037464, 0.17698]$.

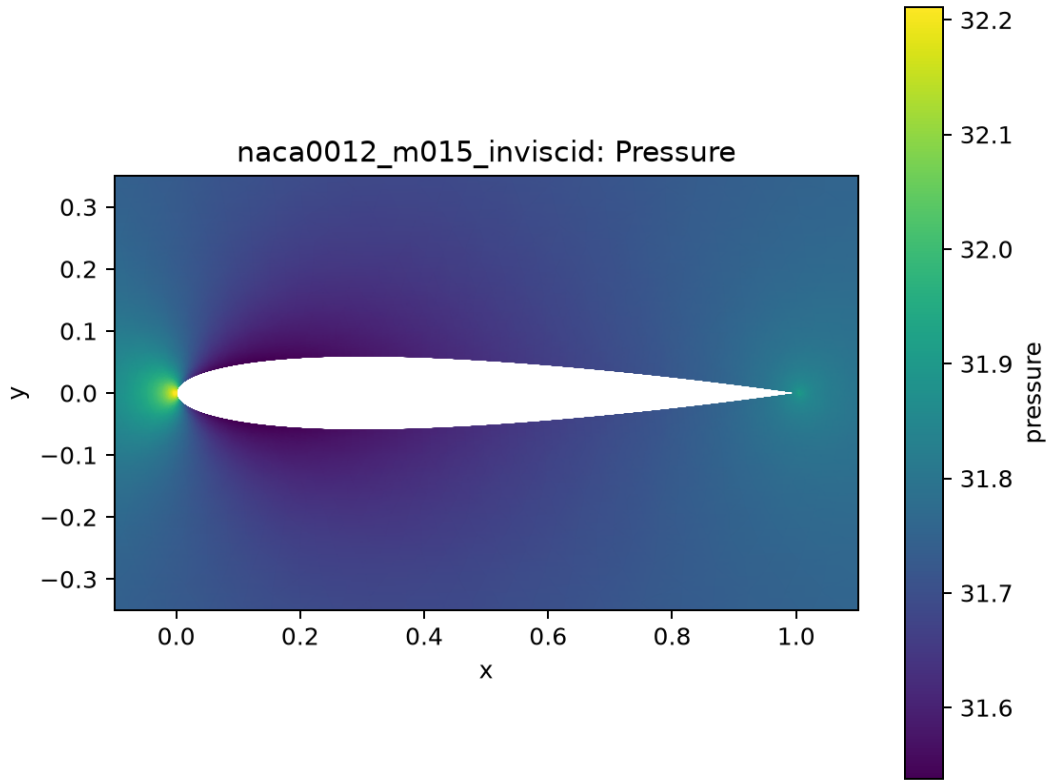


Figure 5: Pressure from the final submitted field; the displayed percentile-clipped range is $[31.538, 32.211]$.

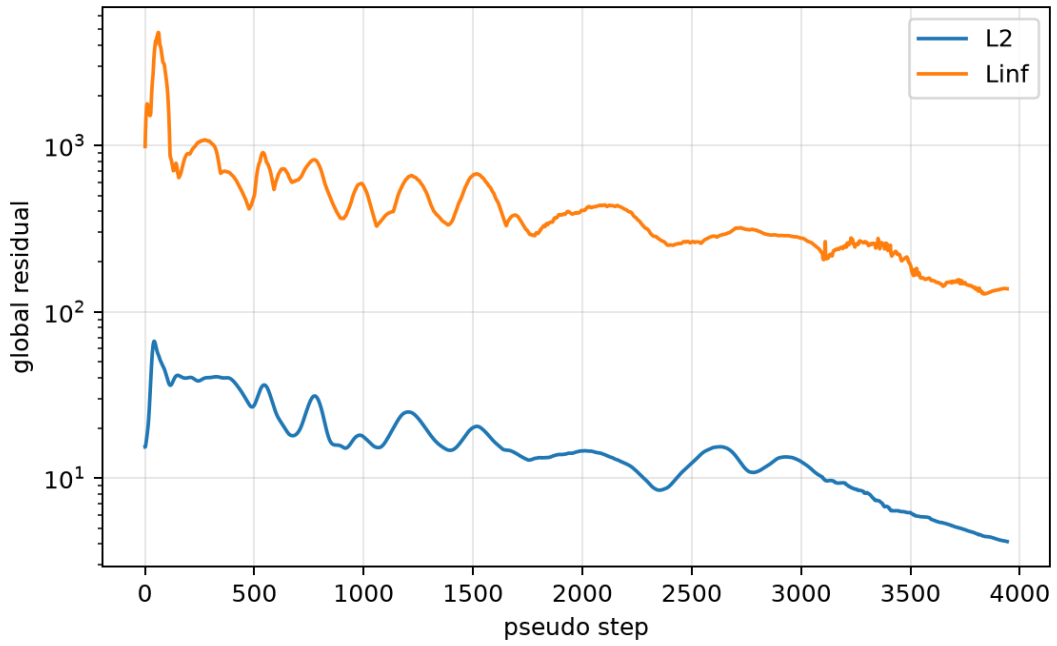


Figure 6: Global residual history.

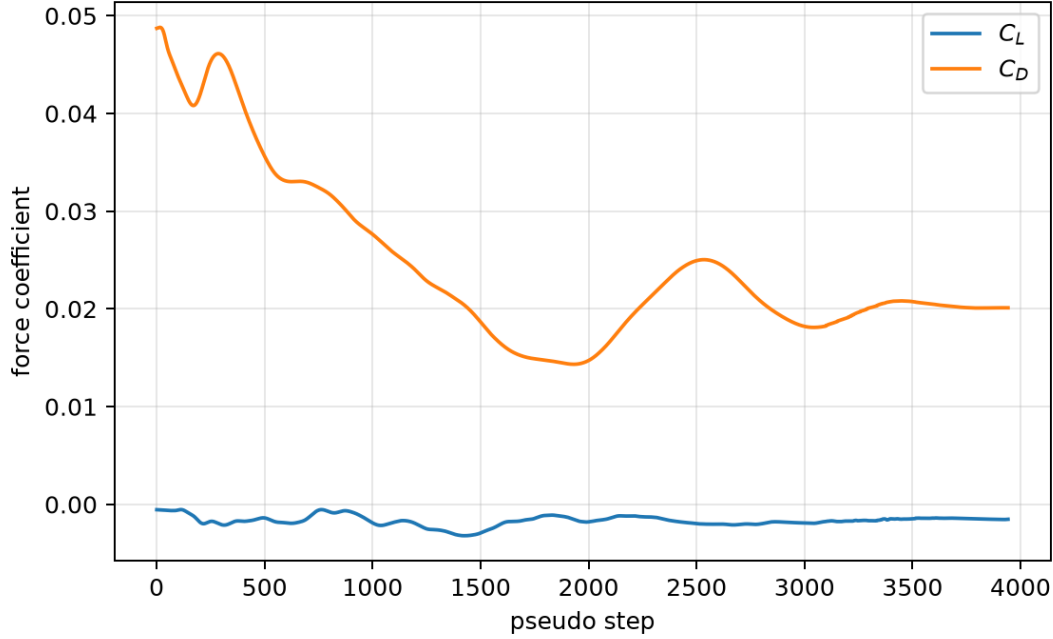


Figure 7: Force-coefficient history.

7.7 naca0012_m200_laminar_re5000

The submitted status is **converged**; the computed first-to-last L_2/L_∞ residual reductions are 3.65/3.56 orders. The final force row gives $C_D = 0.11573$, $C_L = 0.00013323$, and $C_m = -4.9751e - 05$. Figure 26, Figure 27, Figure 28, Figure 29, Figure 30 provide the residual, force, wall C_p/C_f , Mach, and pressure evidence.

7.8 cylinder_m010_laminar_re20

The submitted status is **converged**; the computed first-to-last L_2/L_∞ residual reductions are 5.01/5.01 orders. The final force row gives $C_D = 2.049$, $C_L = 0.00063295$, and $C_m = 2.8194e - 06$. Figure 31, Figure 32, Figure 33, Figure 34, Figure 35 provide the residual, force, wall C_p/C_f , Mach, and pressure evidence.

7.9 cylinder_m010_laminar_re200

The submitted status is **statistically-periodic**; the computed first-to-last L_2/L_∞ residual reductions are 1.29/1.19 orders. The final force row gives $C_D = 1.2747$, $C_L = -0.52747$, and $C_m = 0.0044818$. Figure 36, Figure 37, Figure 38, Figure 39, Figure 40 provide the residual, force, wall C_p/C_f , Mach, and pressure evidence. The final 20% physical-time window is 240.01–300; it gives mean $C_D = 1.2704$, lift amplitude 0.5694, and dominant lift frequency 0.18333 (Strouhal number is identical because the supplied $U_\infty = L_{ref} = 1$). The native force cadence is $\Delta t = 0.01$, spectral resolution is 0.016667, and peak/second magnitude ratio is 1.5571. Figure 41 is the post-transient wake visualization and Figure 42 is the traceable lift spectrum.

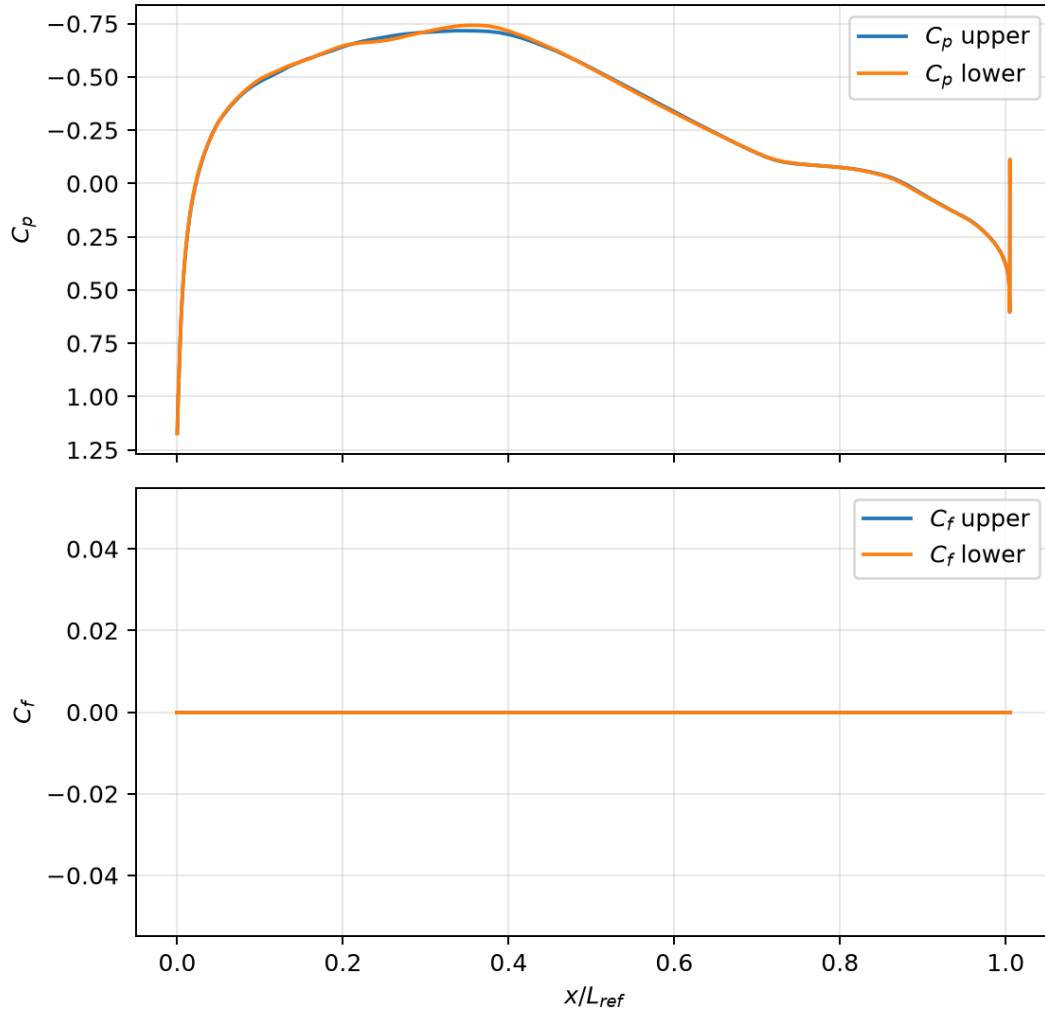


Figure 8: Submitted surface pressure and skin-friction coefficients.

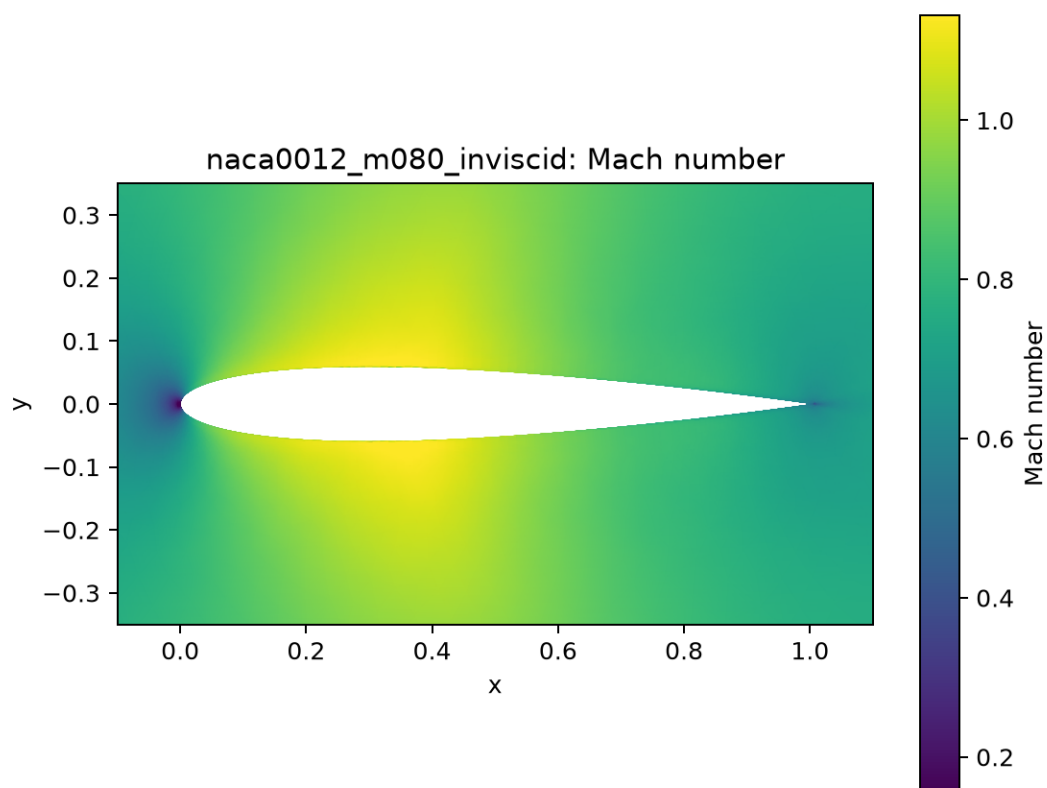


Figure 9: Mach number from the final submitted field; the displayed percentile-clipped range is $[0.15511, 1.1321]$.

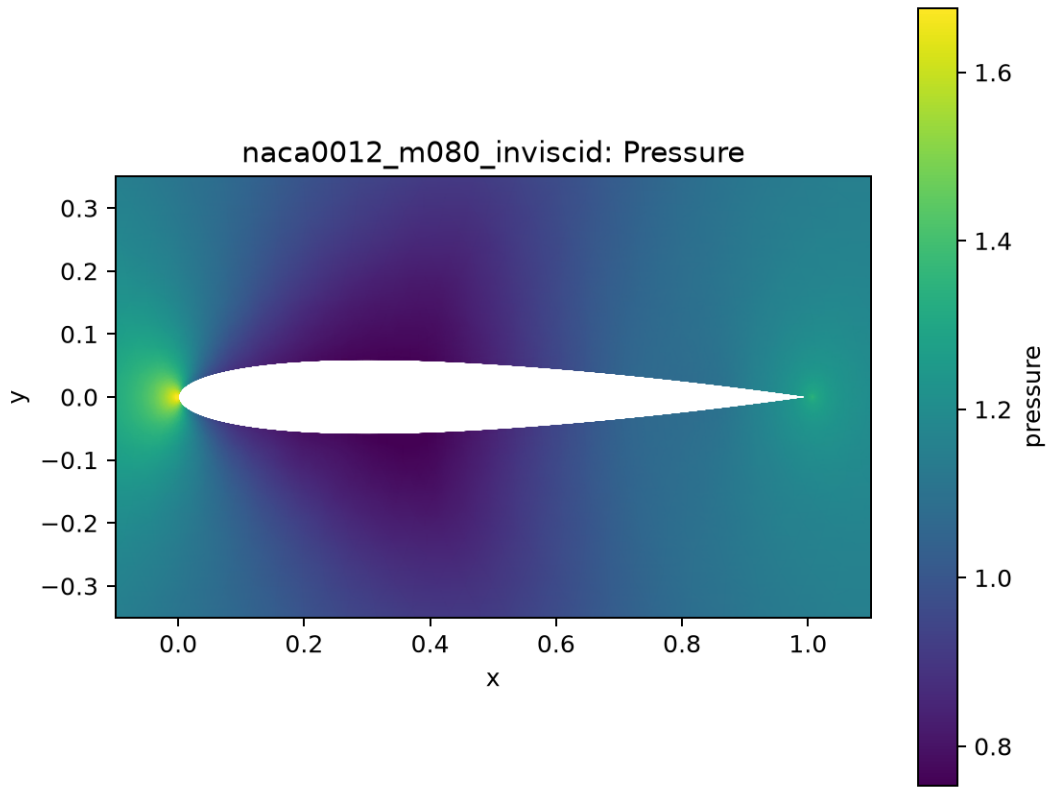


Figure 10: Pressure from the final submitted field; the displayed percentile-clipped range is $[0.75317, 1.676]$.

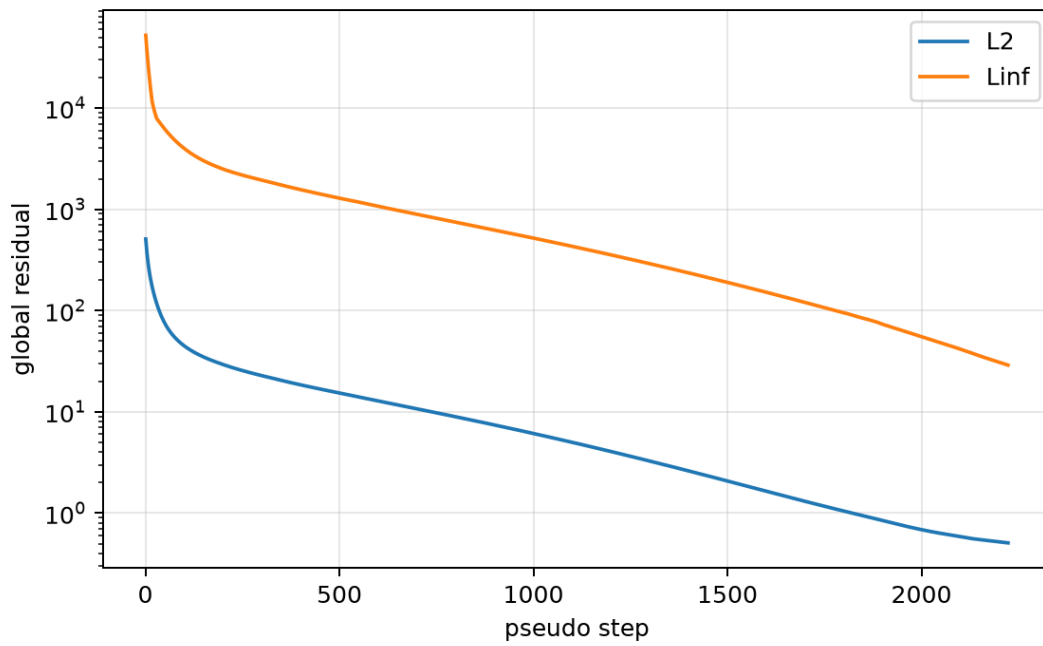


Figure 11: Global residual history.

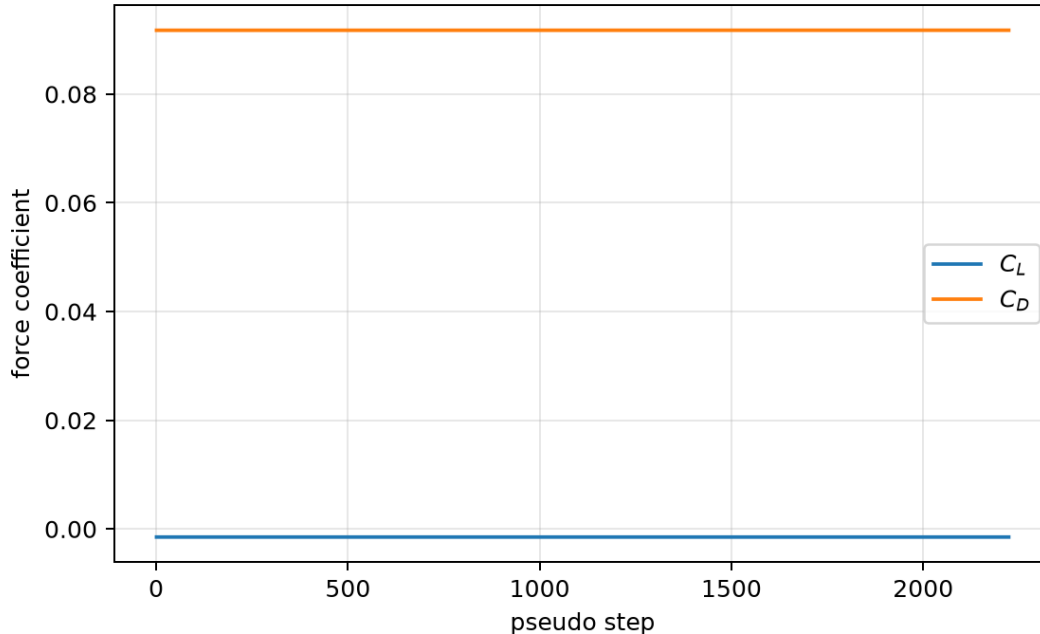


Figure 12: Force-coefficient history.

8 Parallel validation

The following independent completed result packages compare final forces, residual reduction, rank-local load balance, neighbour count, and mean halo send/receive cells across MPI rank counts. Absolute differences are measured against the smallest submitted rank count for the same case. Timing is wall-clock time; the available CSV diagnostics quantify topology and payload but do not independently measure communication time.

Table 7: Rank-count consistency, residual reduction difference, and rank-local communication evidence.

Case	ranks	wall s	C_D	C_L	$ \Delta C_D $	$ \Delta C_L $	$ \Delta R $	balance	nbr mean	send/recv mean
cylinder_m010_laminar_re20	1	822.3	2.049	0.00063295	0	0	0	1	0	0/0
cylinder_m010_laminar_re20	8	106.8	2.049	0.00063295	4e-15	5.95e-14	8.88e-16	1	4.5	121/121
naca0012_m015_inviscid	1	1695	0.00058177	0.00016455	0	0	0	1	0	0/0
naca0012_m015_inviscid	8	209.1	0.00058177	0.00016455	7.47e-15	4.35e-14	5.16e-11	1.01	3.5	112/112

9 Limitations and failure analysis

This report is an evidence renderer, not a numerical adjudicator. It rejects failed statuses and failed contract-derived sanity checks, but it cannot prove grid convergence, model-form accuracy, or source-code behavior from output alone. Any remaining limitations are retained verbatim in the submitted run-manifest notes and should be evaluated alongside the residual, force, surface, field, and rank-comparison evidence above.

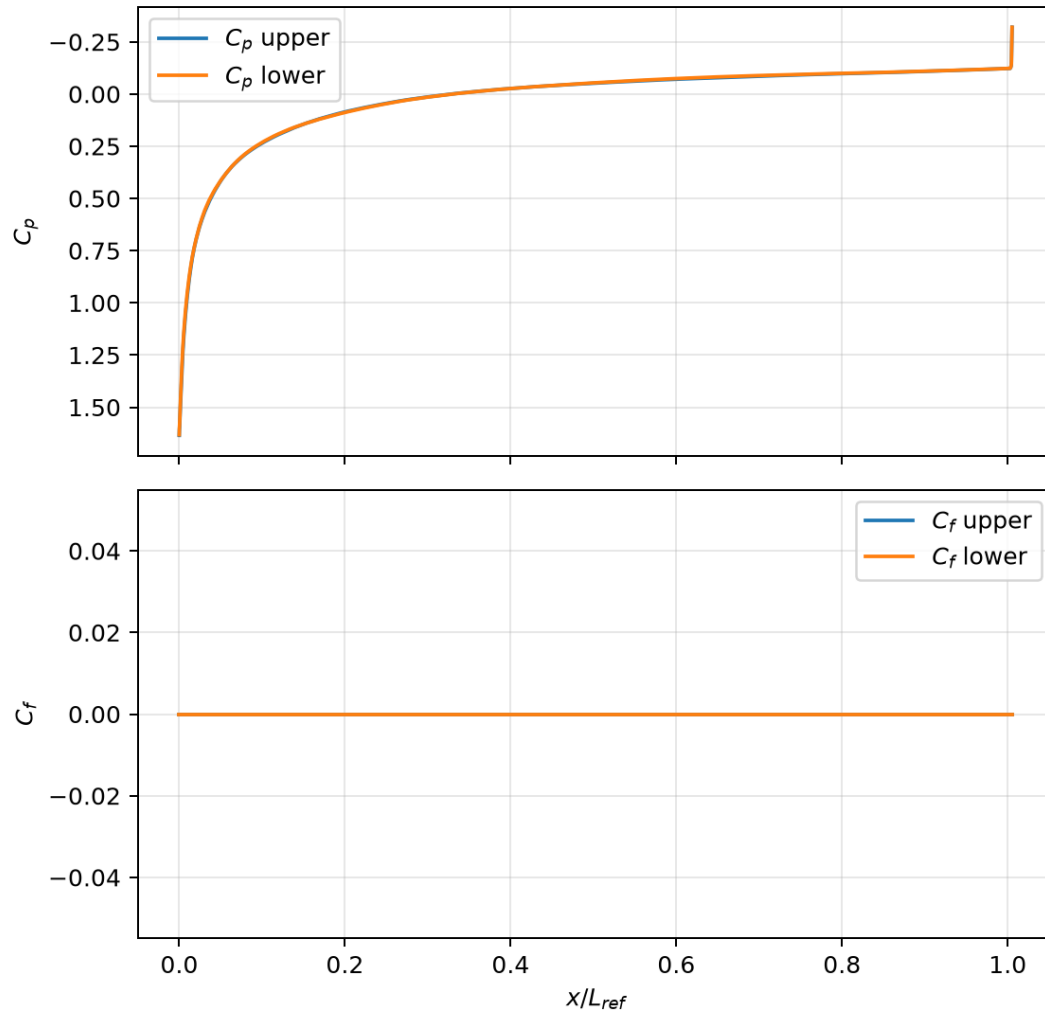


Figure 13: Submitted surface pressure and skin-friction coefficients.

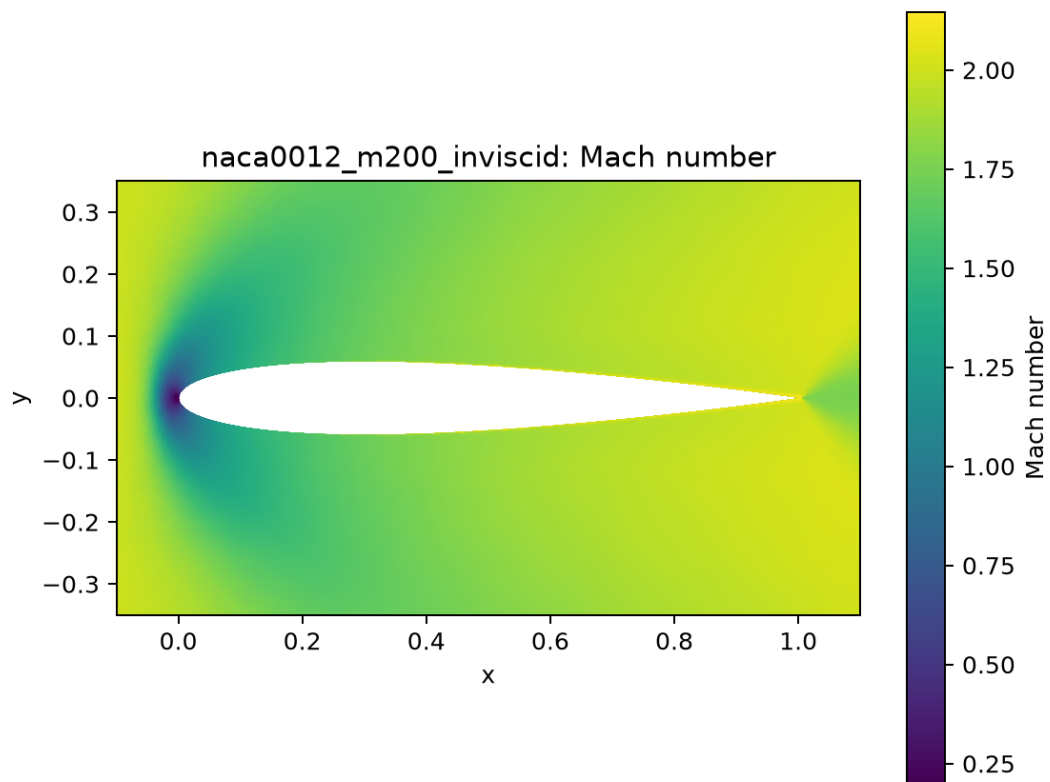


Figure 14: Mach number from the final submitted field; the displayed percentile-clipped range is $[0.19879, 2.1459]$.

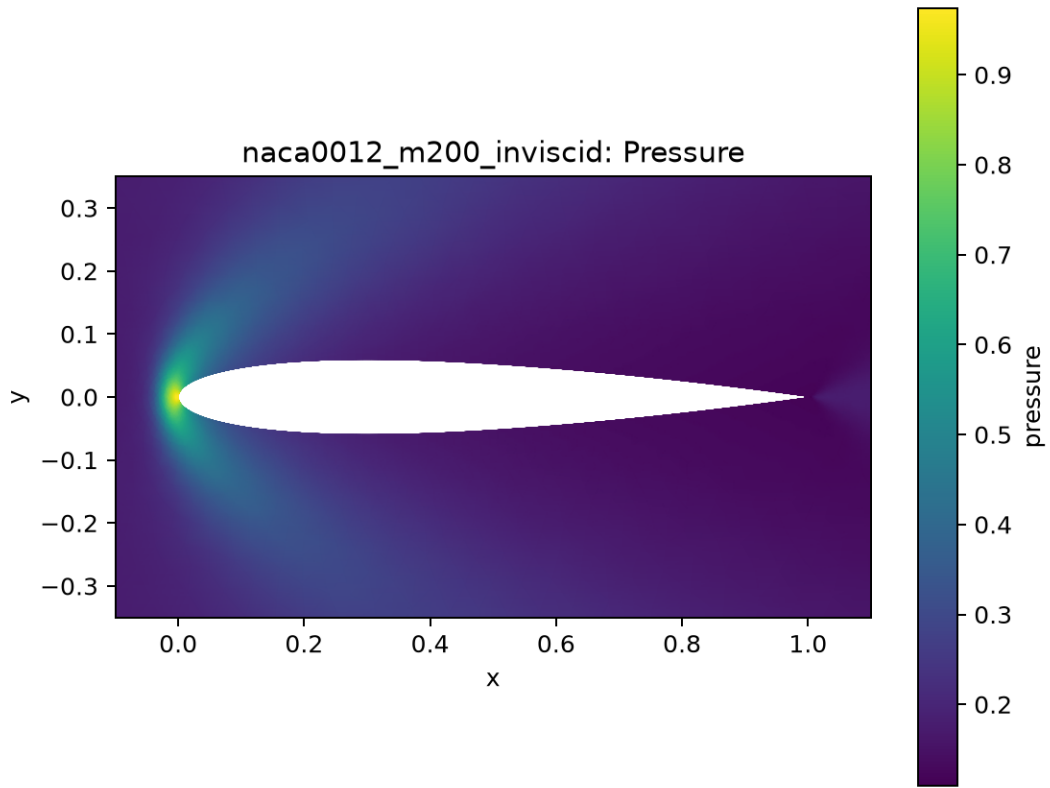


Figure 15: Pressure from the final submitted field; the displayed percentile-clipped range is $[0.10953, 0.97393]$.

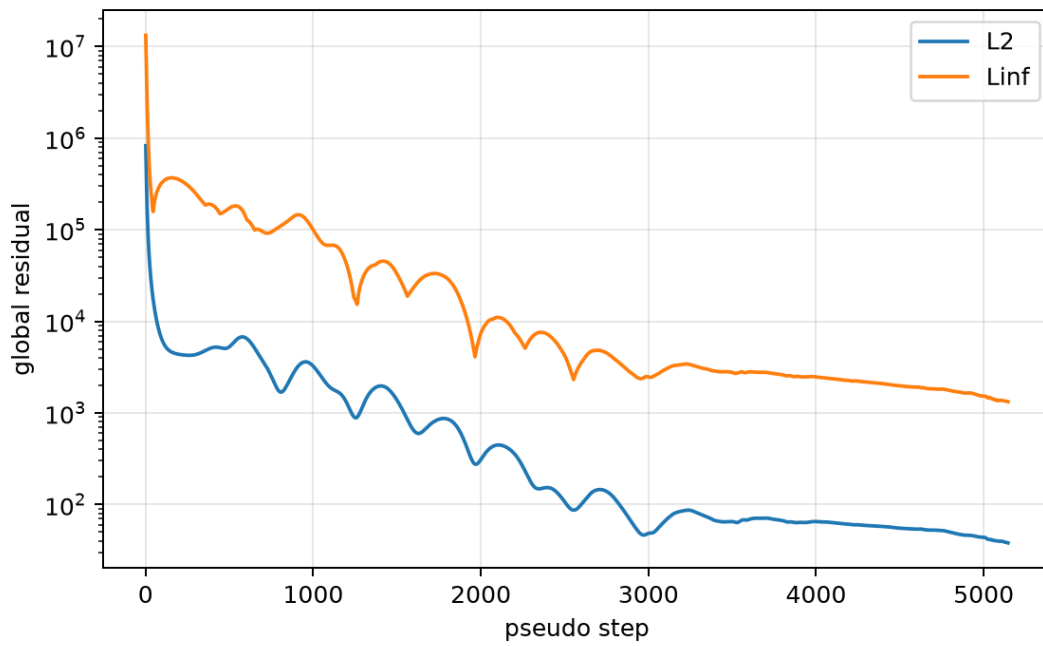


Figure 16: Global residual history.

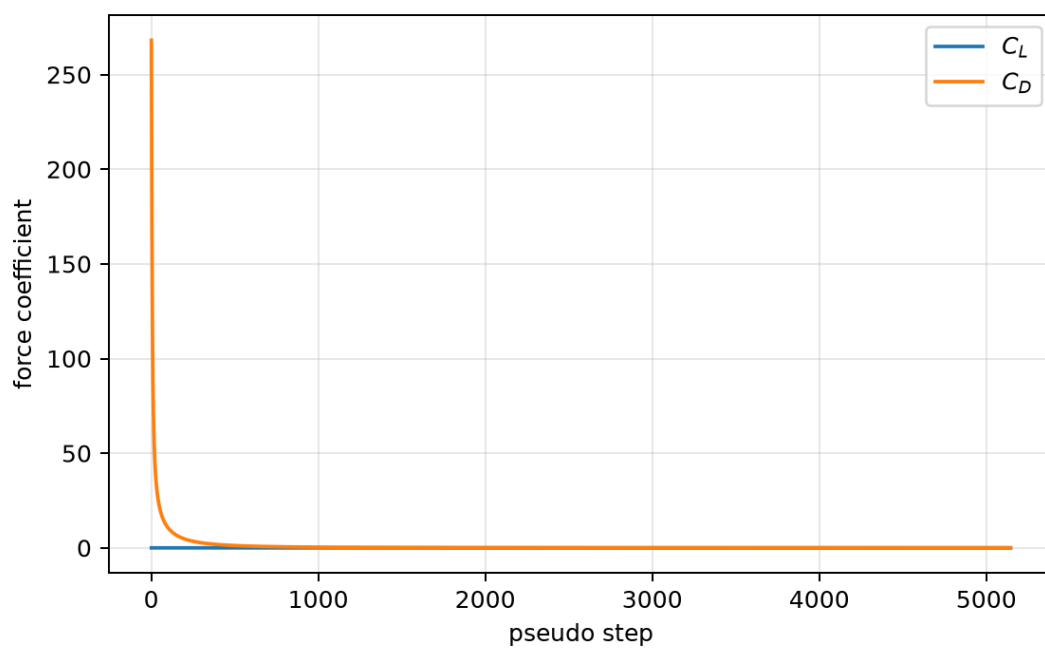


Figure 17: Force-coefficient history.

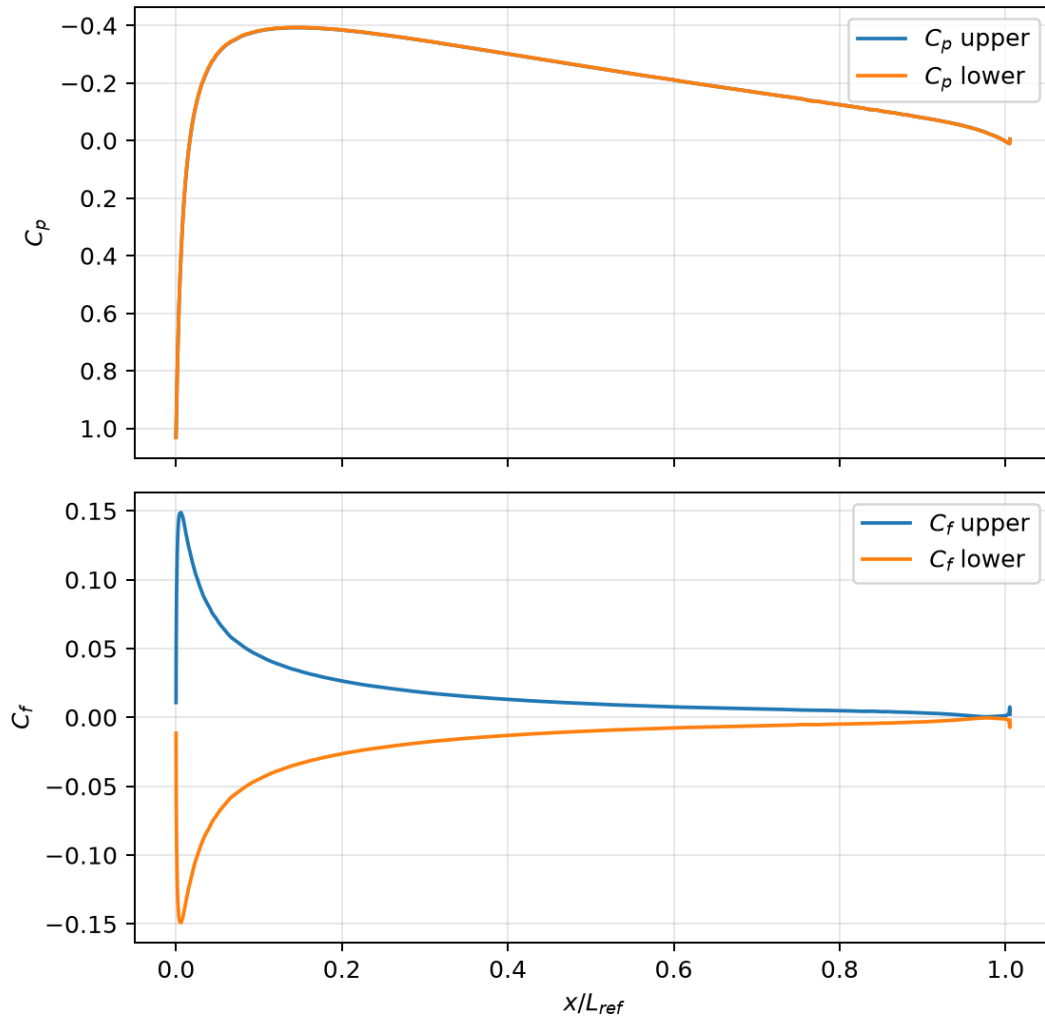


Figure 18: Submitted surface pressure and skin-friction coefficients.

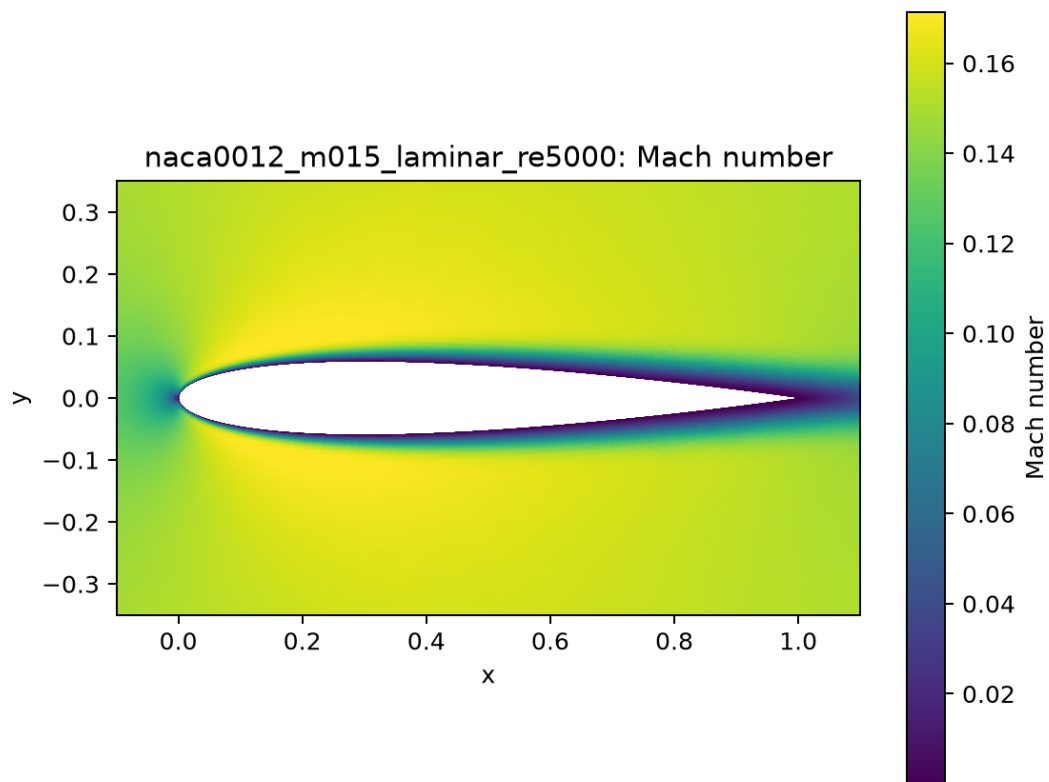


Figure 19: Mach number from the final submitted field; the displayed percentile-clipped range is $[3.4982\text{e-}06, 0.17134]$.

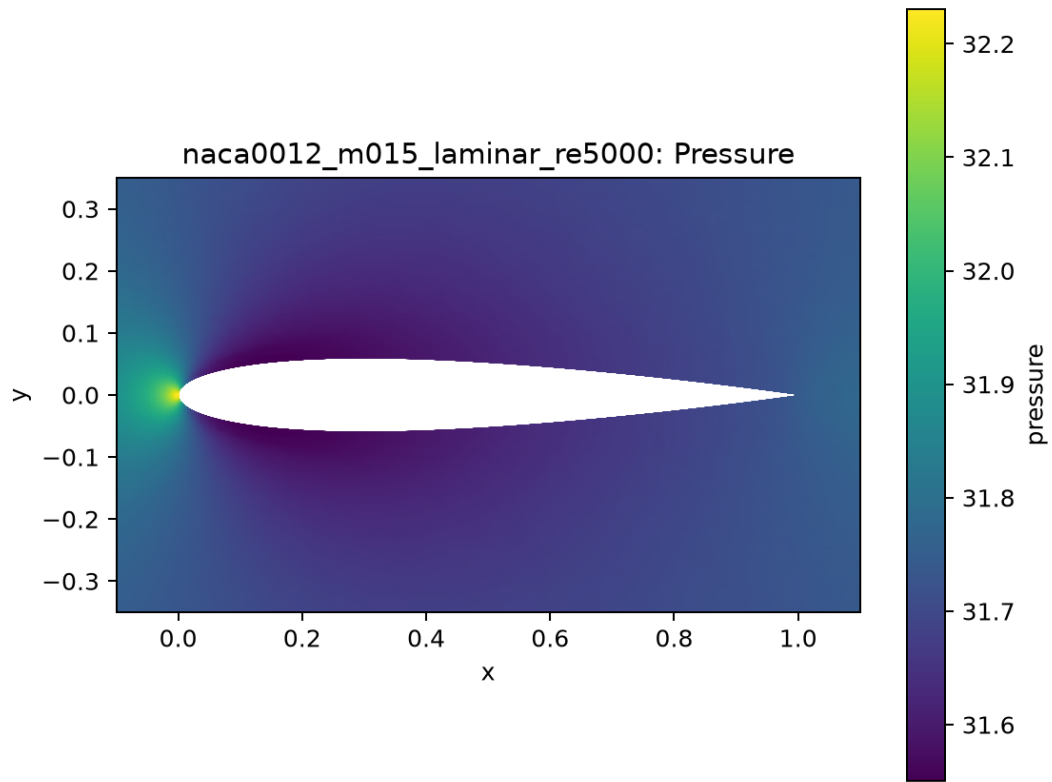


Figure 20: Pressure from the final submitted field; the displayed percentile-clipped range is [31.55, 32.231].

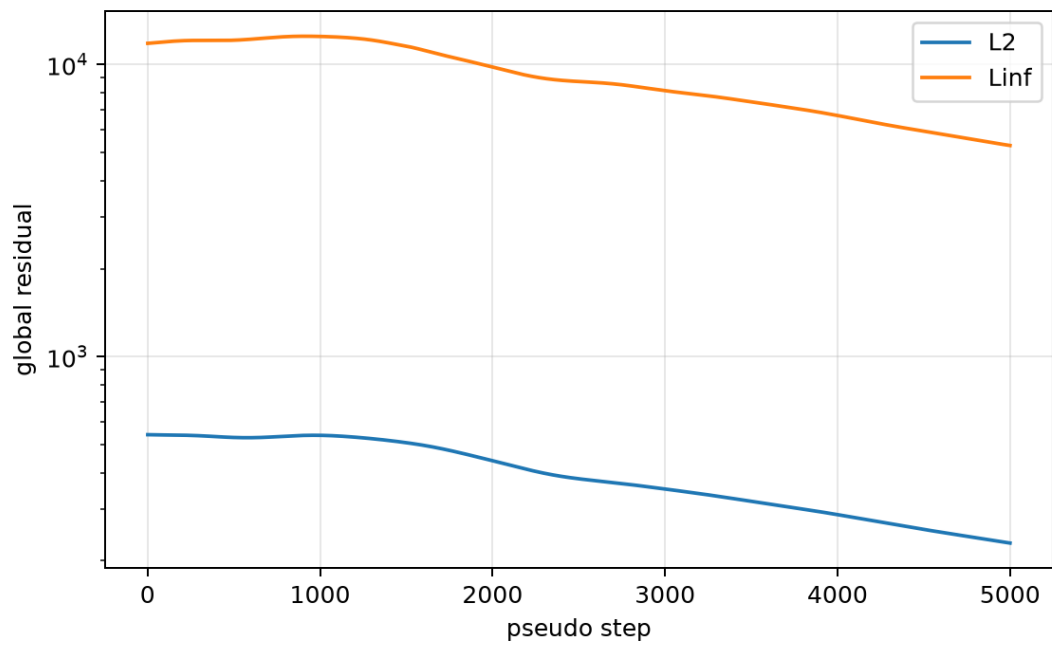


Figure 21: Global residual history.

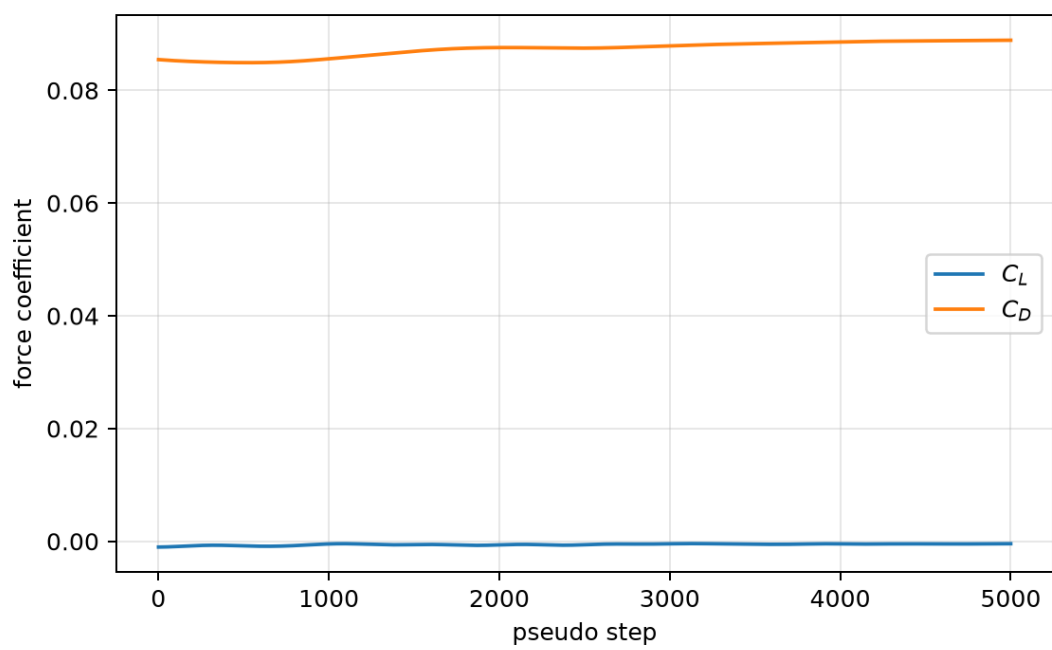


Figure 22: Force-coefficient history.

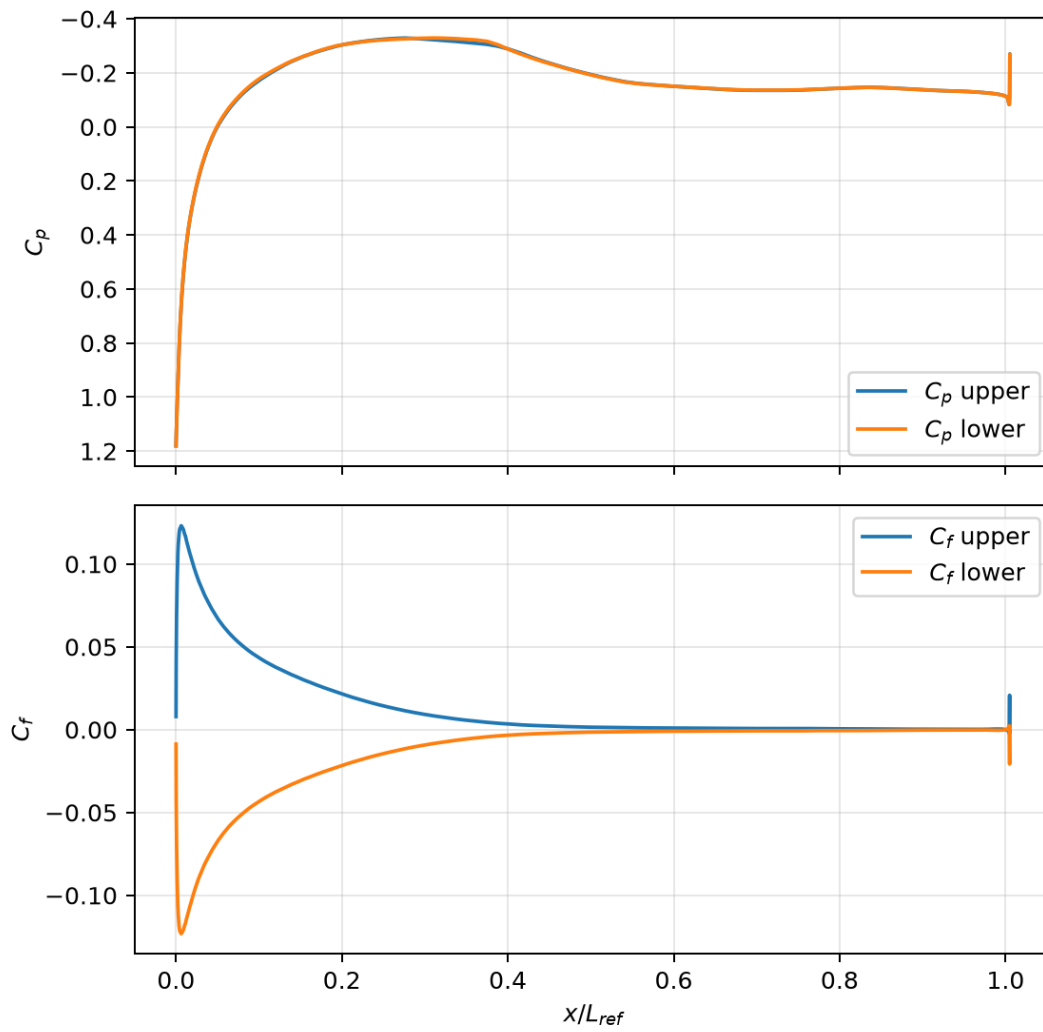


Figure 23: Submitted surface pressure and skin-friction coefficients.

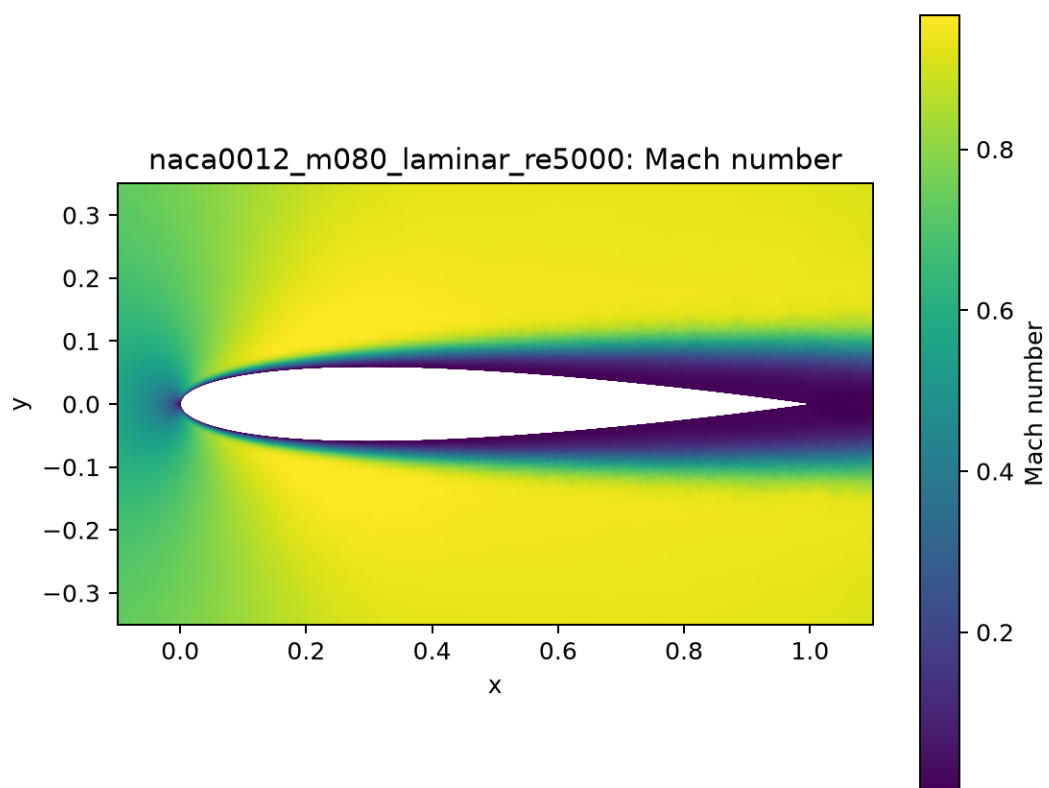


Figure 24: Mach number from the final submitted field; the displayed percentile-clipped range is $[0.00053447, 0.96641]$.

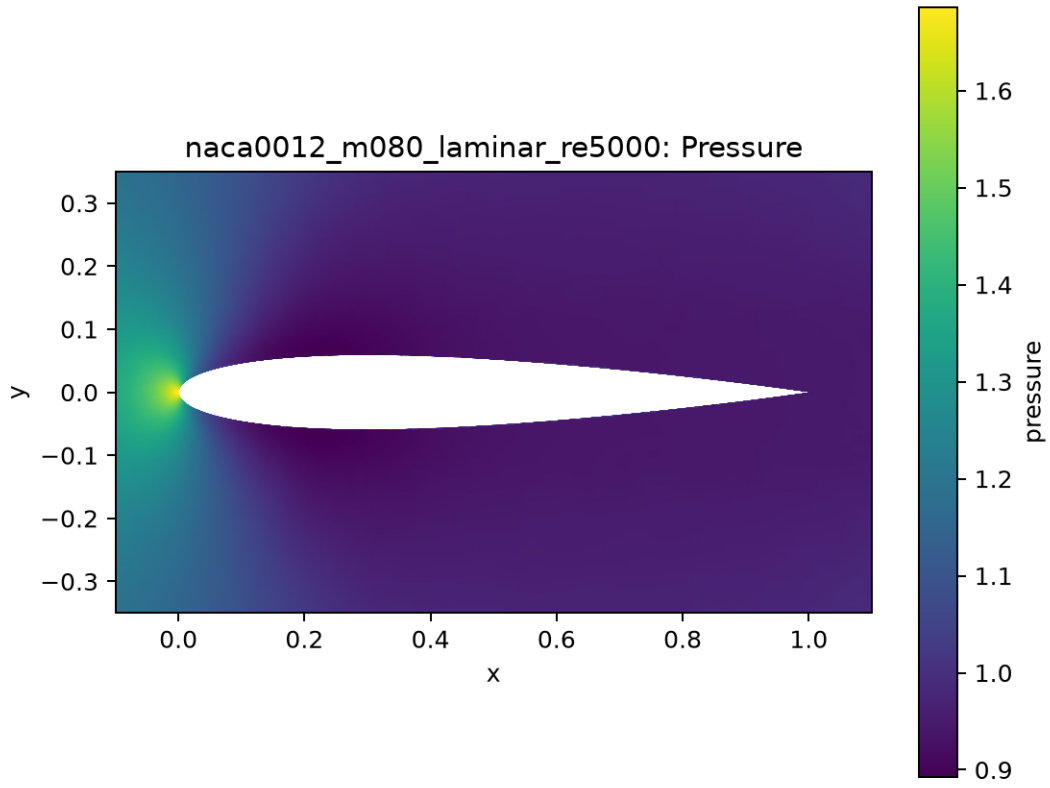


Figure 25: Pressure from the final submitted field; the displayed percentile-clipped range is $[0.89265, 1.6857]$.

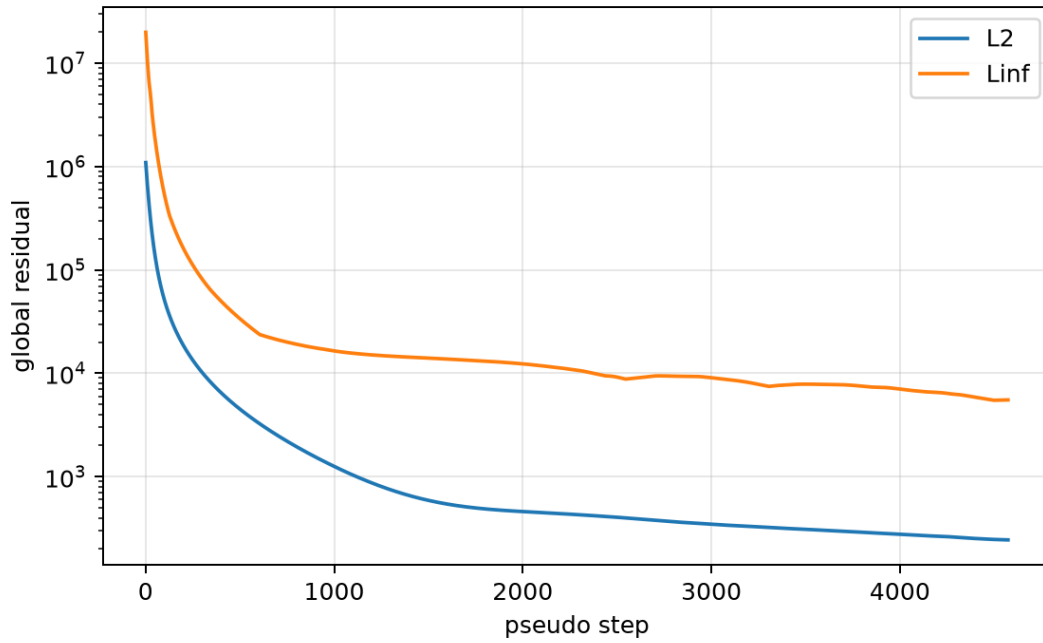


Figure 26: Global residual history.

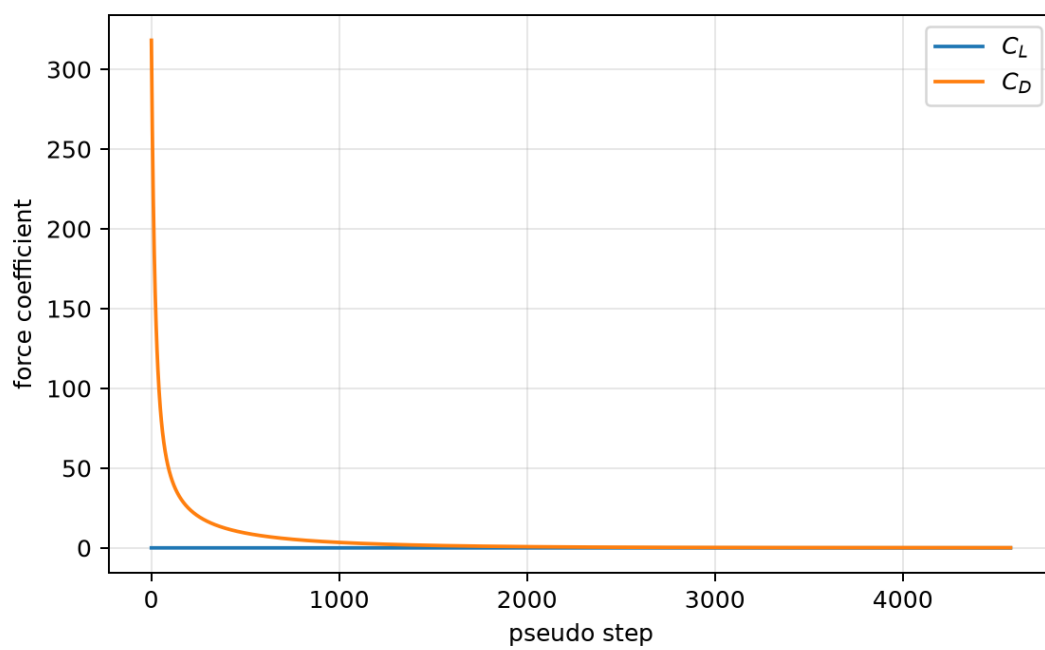


Figure 27: Force-coefficient history.

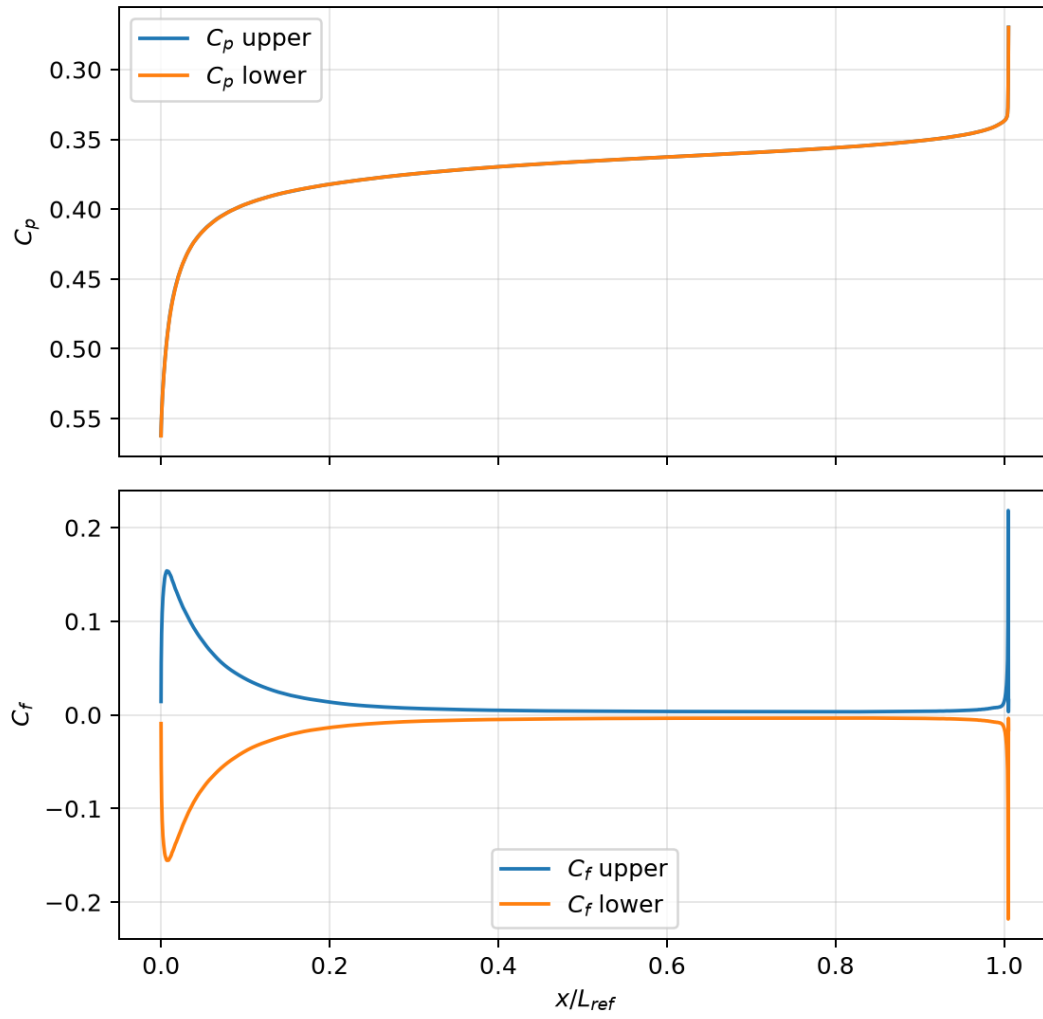


Figure 28: Submitted surface pressure and skin-friction coefficients.

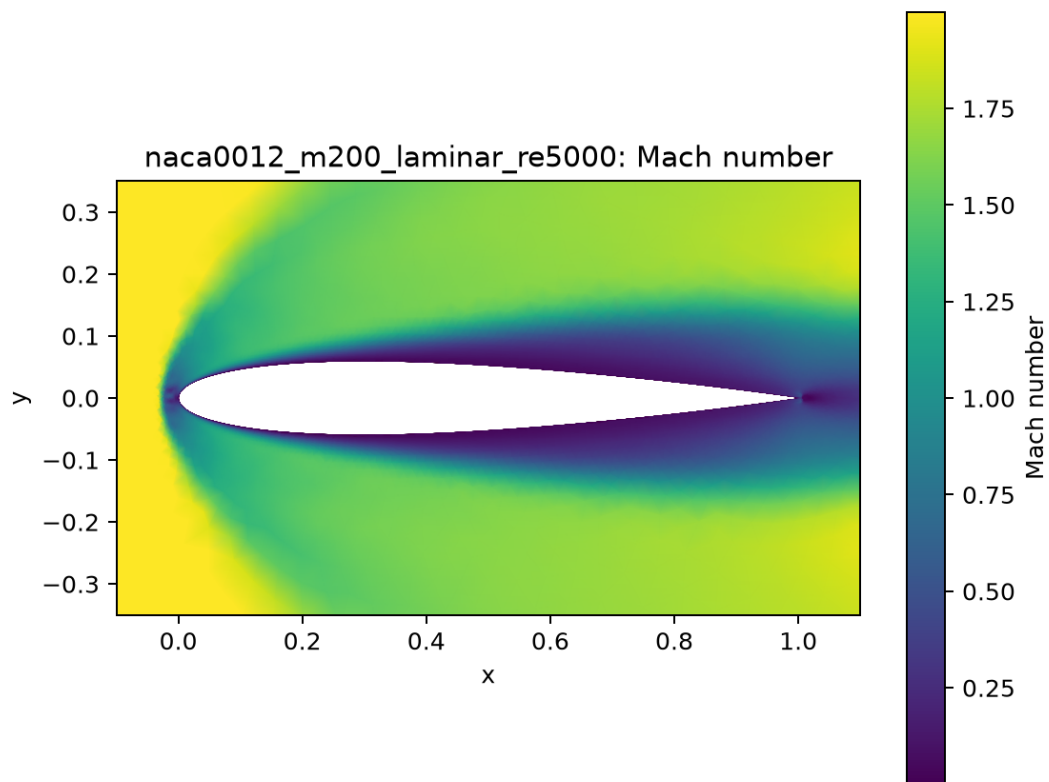


Figure 29: Mach number from the final submitted field; the displayed percentile-clipped range is $[0.0013443, 1.9993]$.

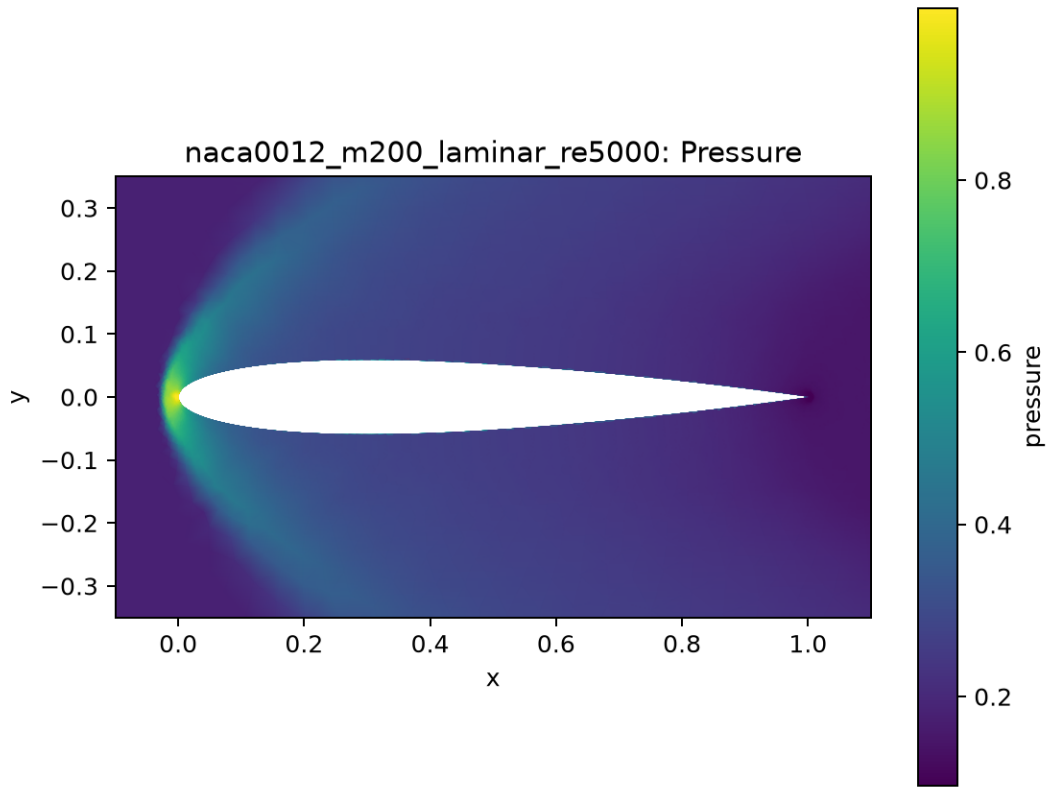


Figure 30: Pressure from the final submitted field; the displayed percentile-clipped range is $[0.096422, 0.99969]$.

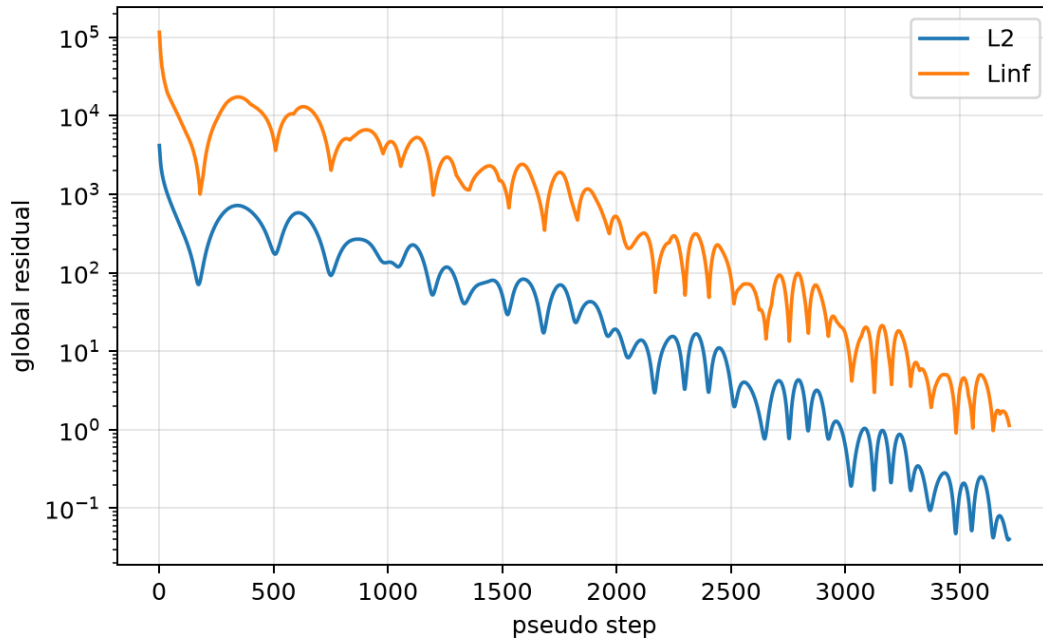


Figure 31: Global residual history.

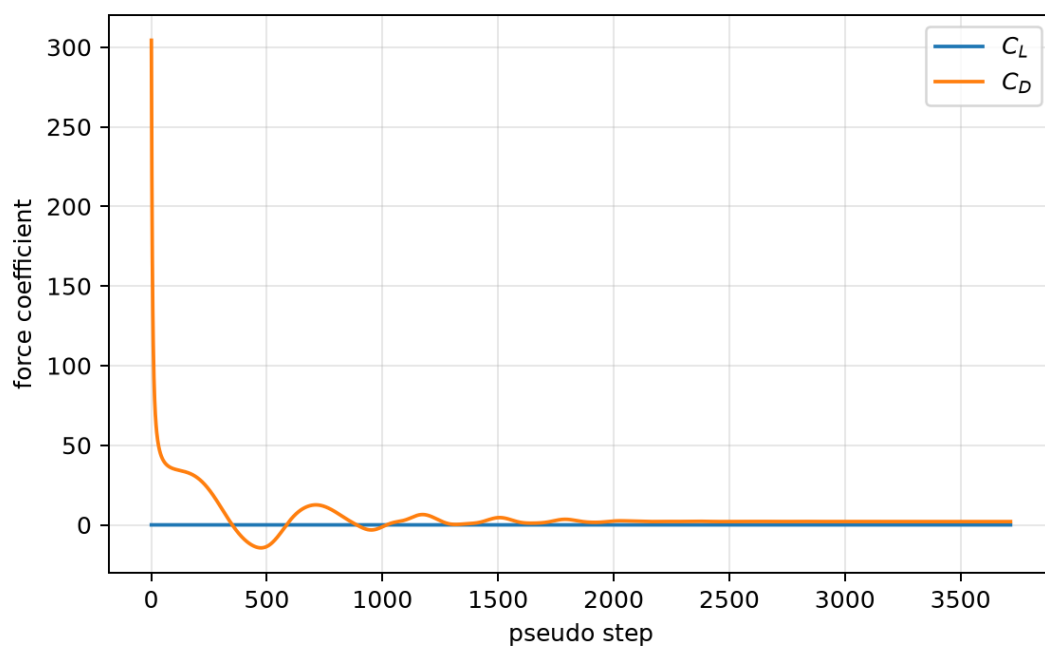


Figure 32: Force-coefficient history.

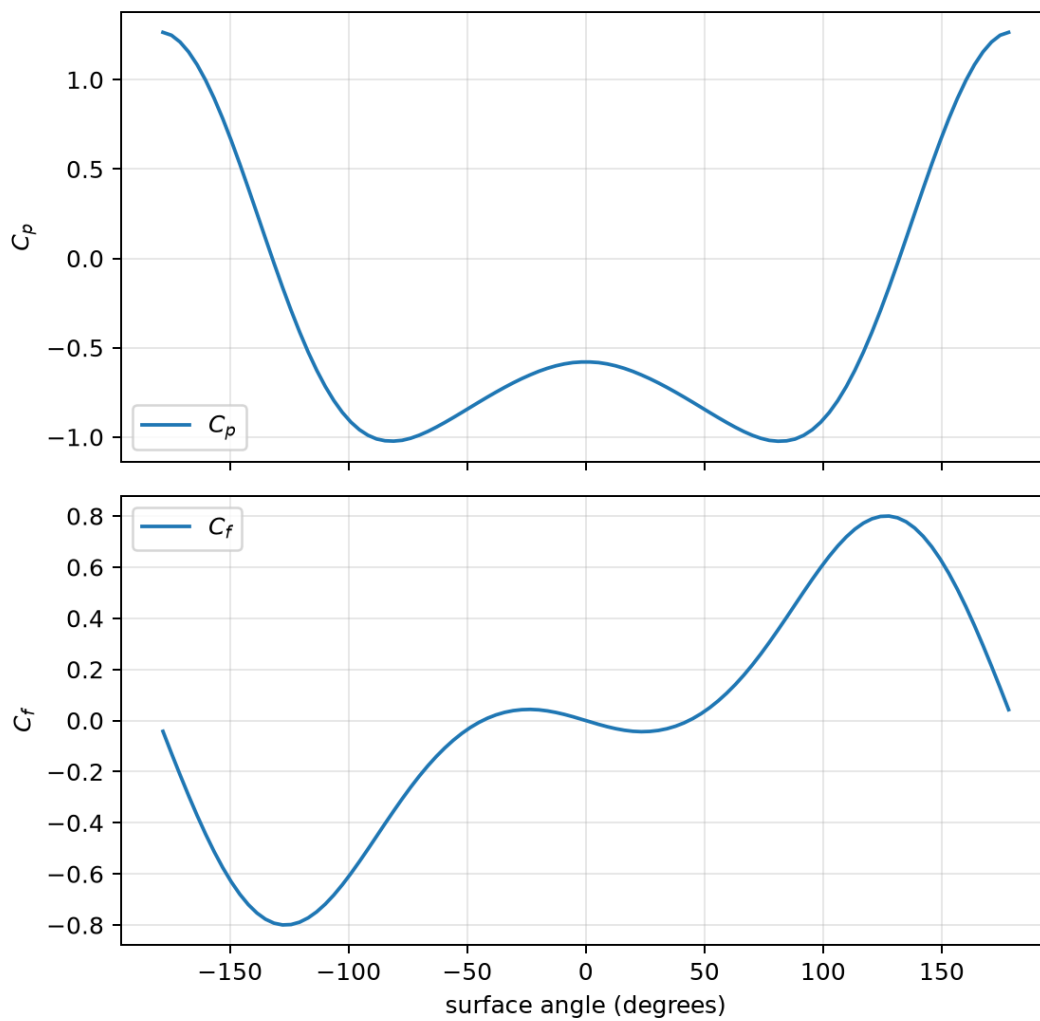


Figure 33: Submitted surface pressure and skin-friction coefficients.

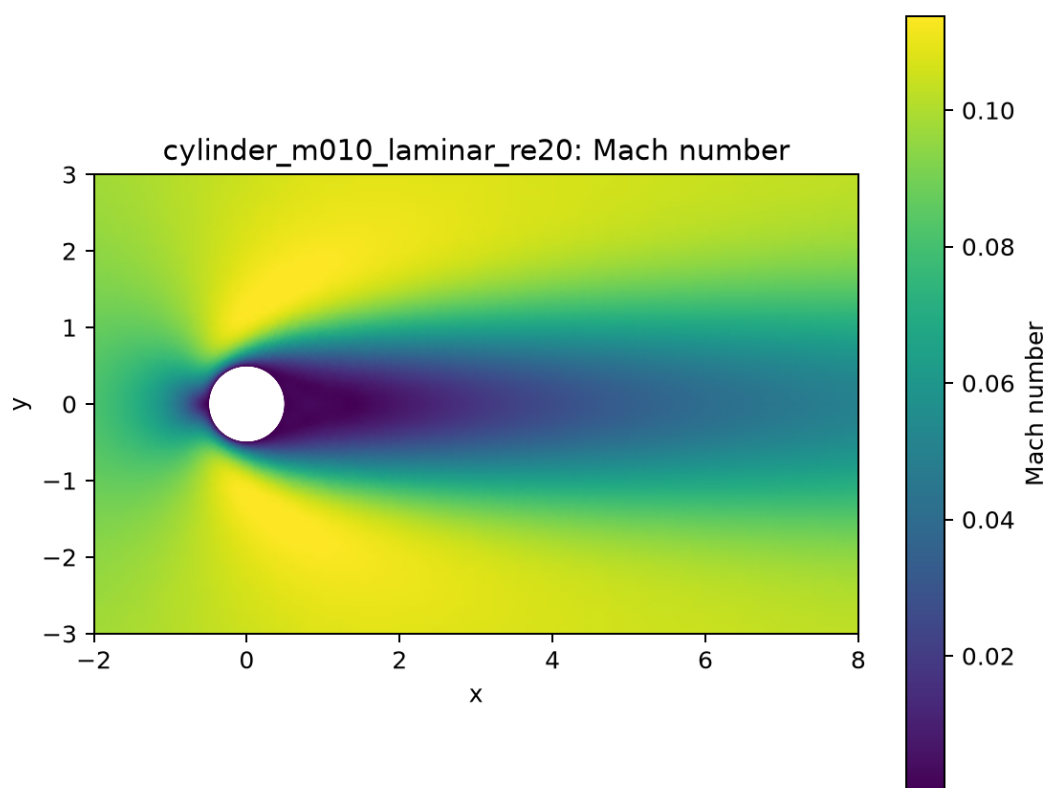


Figure 34: Mach number from the final submitted field; the displayed percentile-clipped range is $[8.4008\text{e-}05, 0.11378]$.

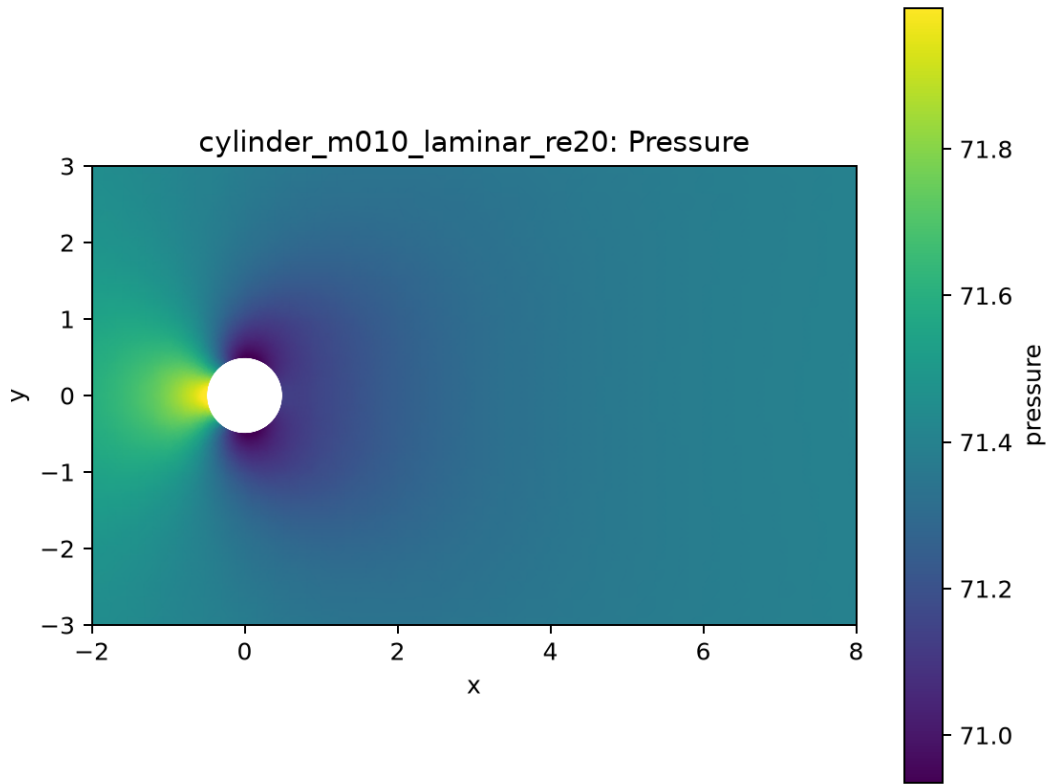


Figure 35: Pressure from the final submitted field; the displayed percentile-clipped range is [70.935, 71.992].

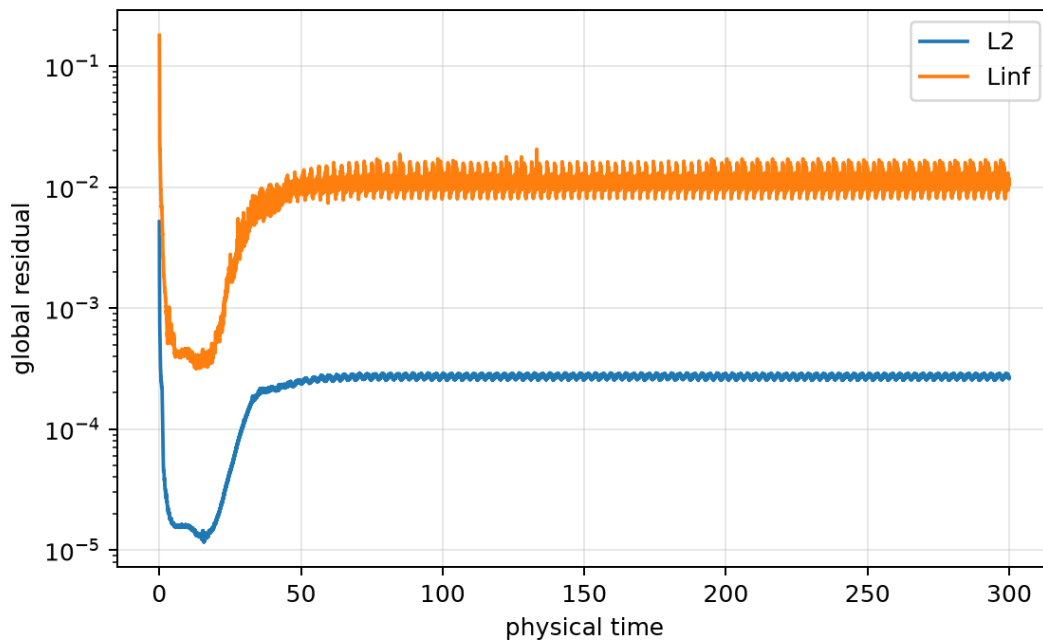


Figure 36: Global residual history.

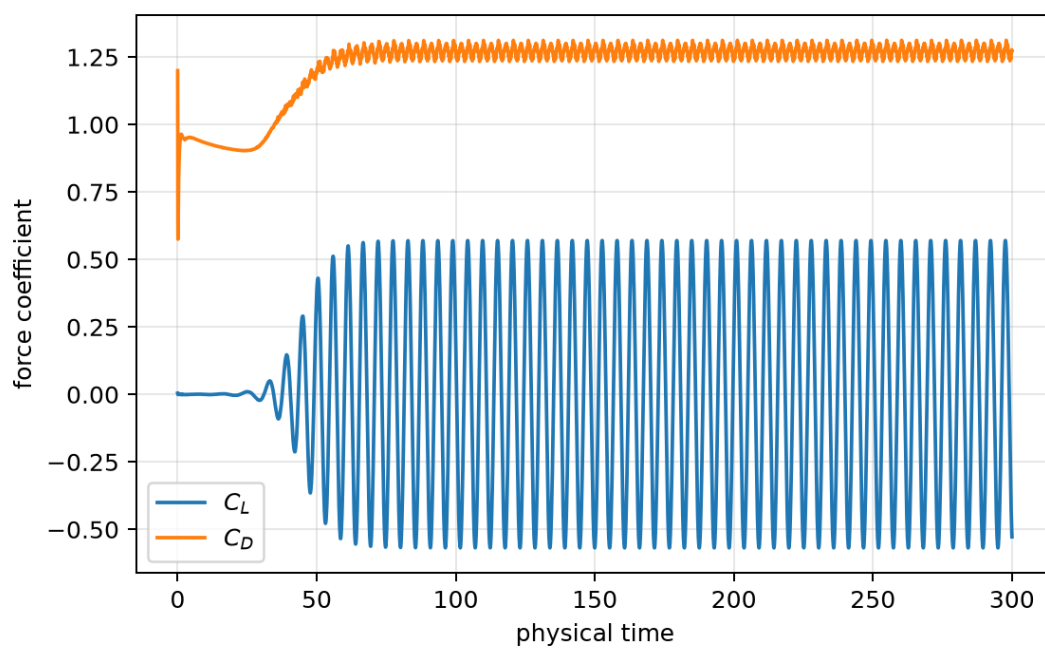


Figure 37: Force-coefficient history.

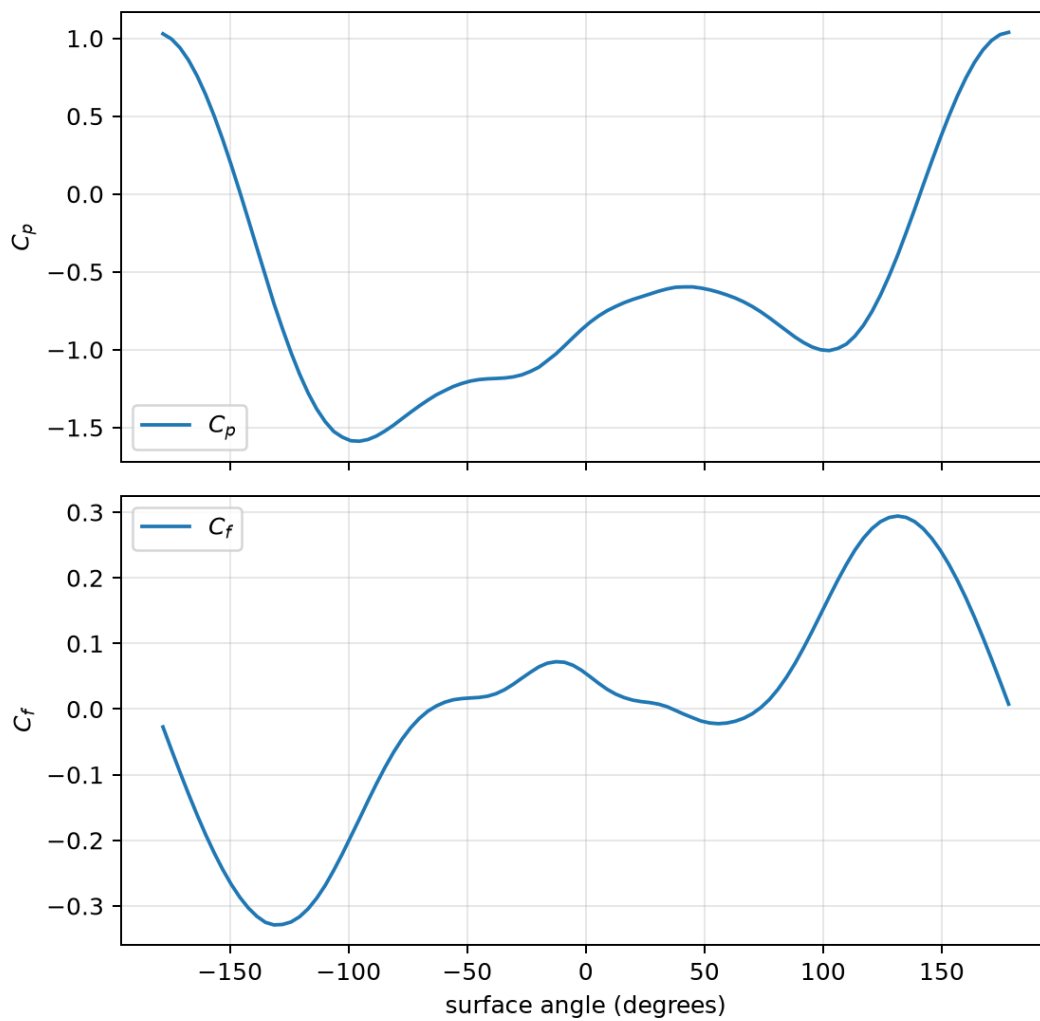


Figure 38: Submitted surface pressure and skin-friction coefficients.

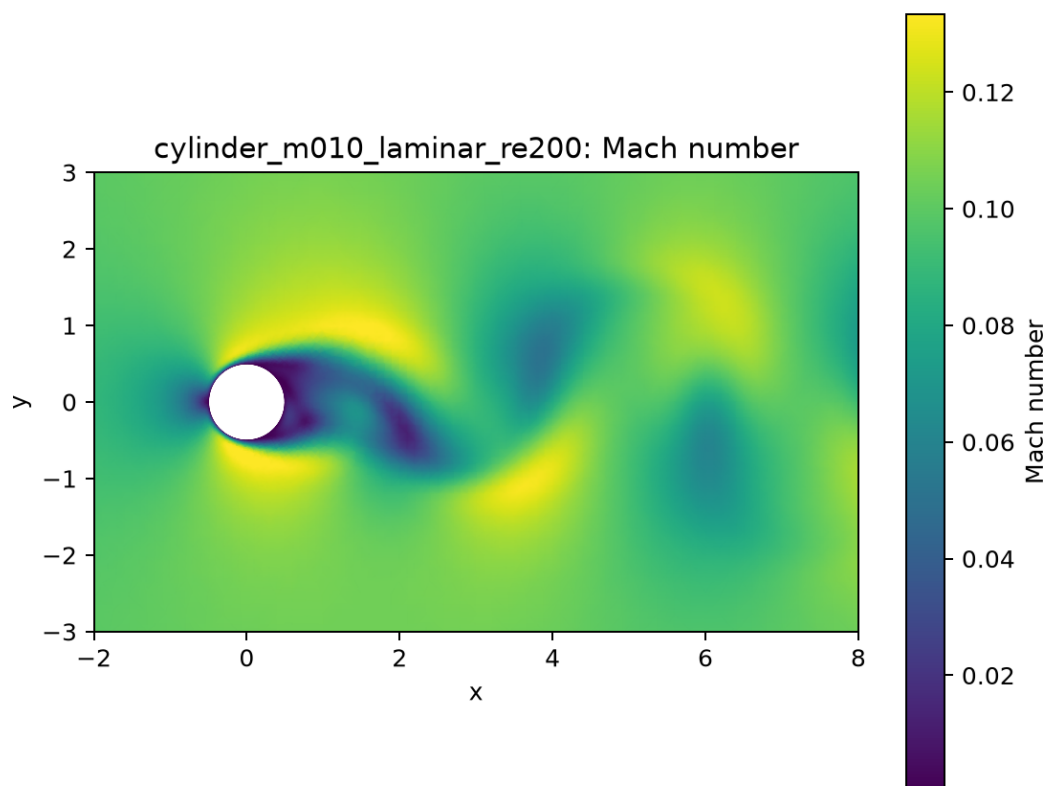


Figure 39: Mach number from the final submitted field; the displayed percentile-clipped range is $[0.00037344, 0.13335]$.

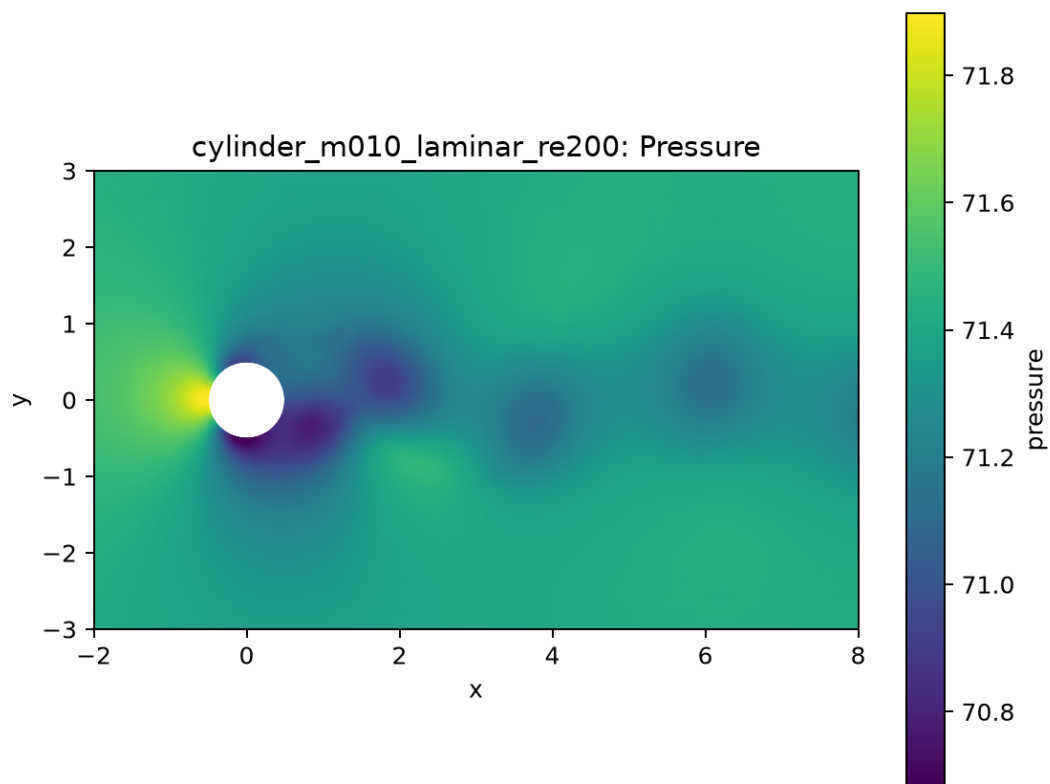


Figure 40: Pressure from the final submitted field; the displayed percentile-clipped range is [70.682, 71.898].

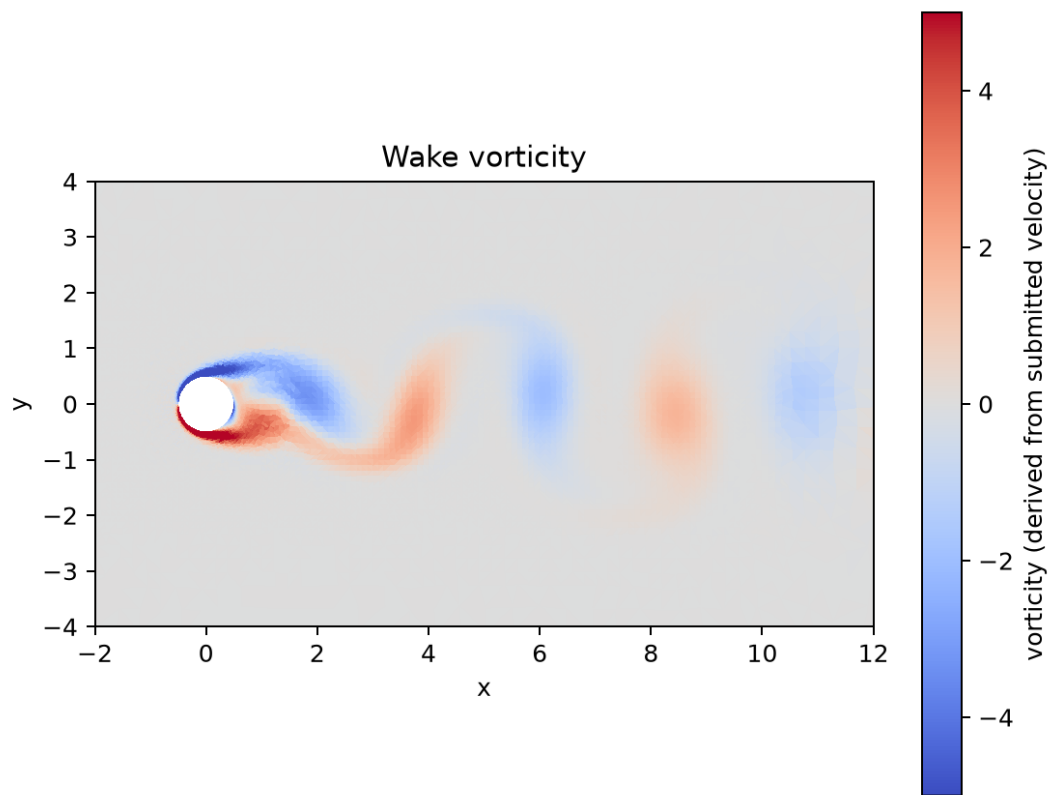


Figure 41: Vorticity derived from final submitted velocity components; symmetric colour clipping is $[-5, 5]$.

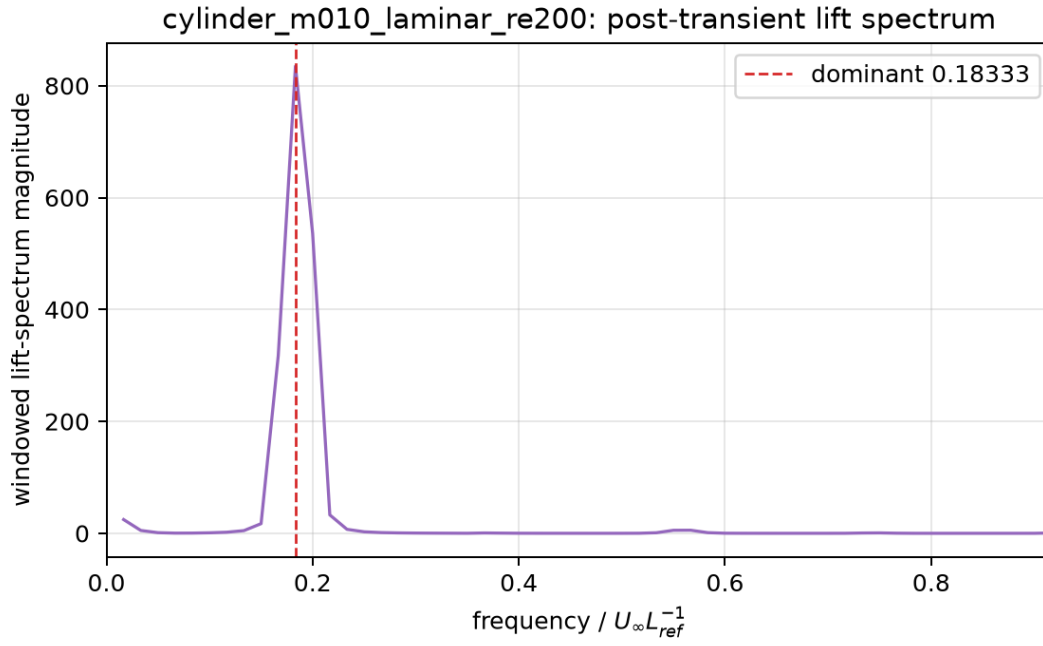


Figure 42: Post-transient detrended Hann-window lift spectrum from forces.csv; sample interval 0.01, window 59.99, frequency resolution 0.016667, dominant frequency 0.18333, peak/second ratio 1.56. The displayed low-frequency window ends at 0.91667; the FFT was evaluated through the Nyquist frequency 50.